

# MIDAS GUI Documentation

## MIDAS GUI — User Documentation

**Version:** 1.0.0 **Application:** `midas-gui` (or `python -m midas_gui`) **Backends:** `midas_calibrate_v2`, `midas_integrate_v2`, `midas_calibrate`, `midas_hkls`, `midas_distortion` **Last updated:** 2026-08-25 (Open Project’s “Populate from project” now redraws the recorded run’s **results**, not just the input fields: Calibrate’s predicted-ring overlay, Radial Profile, and Eta vs R Cake reappear immediately (rings need only the stored geometry; profile/cake also need the run’s data file to still be loadable), and Batch Integrate’s Waterfall/Stacked-profiles views are replayed from the attempt’s saved per-frame data — in Hydra mode, independently per GE panel, so switching the GE1–GE4 toolbar shows that panel’s own recorded run. A new bold, high-contrast **Project:** ... label also appears at the far right of the tab-bar header row whenever a project is open (in addition to the existing quiet status-bar one), so an active project is now hard to miss. See §5, §7, and §17 “File ▶ Project (FAIR provenance)”.)

**Previously:** (2026-08-25, Data Viewer and Calibrate: loading a Data file now auto-populates **pixel size** (from the detector encoded in the filename — `.ge1` — `.ge5` → GE, 200  $\mu\text{m}$ ; `.vrx` → Varex, 150  $\mu\text{m}$ ; `.pxrd` is recognized as Pixirad but has no known pixel size to auto-fill) and, for an HDF5 frame file, **wavelength** (from its recorded beam energy, `keV` →  $\text{\AA}$ ). Only active for the **1-ID-E**, **20-ID-D**, and **20-ID-E** profiles — these are beamline-specific filename/metadata conventions. Applies to both single-detector and Hydra mode in both tabs; anything not detected is left at its previous/default value. Batch Integrate is unaffected — it always gets pixel size/wavelength from the calibration it’s handed. See §3 and §5 “Data Loader panel”).

**Previously:** (2026-08-25, Switching profiles from the header **Profile:** [combo ▼] now actually refreshes every dropdown/menu whose choices come from that profile, not only tab visibility and Hydra availability: the Data Viewer’s **Live Data** PV dropdown, the Calibrate tab’s **Calibrant** dropdown (single-detector and every Hydra panel), and the pixel-size-preset / K-edge-foil popup menus all now show the newly-active profile’s values immediately — previously these kept the *old* profile’s list until the app was restarted. Seeded numeric/path defaults (wavelength, pixel size, Lsd, beam-centre, default files) are unchanged — they still only seed a field once when a tab is built, so in-progress edits are never overwritten by a profile switch. See §15 “Profiles”).

**Previously:** (2026-08-25, A prominent **Profile:** [combo ▼] dropdown now sits at the top-left of the main window, in the same row as the tab bar — switching profiles is instant, no more digging into Preferences. The **Hydra** option on Data Viewer/Calibrate/Batch Integrate’s mode ribbon now only appears when the active profile is **1-ID-E** — the only beamline with that detector — falling back to Single detector automatically if you switch away while Hydra mode is active. See §15 “Profiles” and §3 “Mode ribbon”).

**Previously:** (2026-08-24, File ▶ **Open Project...** now offers to *populate* the GUI from the project’s own recorded attempts, not just make it active for future logging — a **Populate from project** dialog lets you pick, per panel (single-detector or each present Hydra GE panel), which recorded Calibrate/Batch Integrate attempt to load. See §17 “File ▶ Project (FAIR provenance)” → “Opening a project can populate the GUI”).

**Previously:** (2026-08-24, Tab 2 — Calibrate: Hydra mode’s **Use manual seed/Feed result back to seed** are now one shared choice linked across all 4 GE panels (seed *values* stay independent per panel); both Single-detector and Hydra Calibrate gain a new **Eta vs R Cake** tab alongside Radial Profile — a 2-D ( $\eta$ , R) intensity heatmap, the standard area-detector “cake” visualization, R on X and  $\eta$  on Y.)

**Previously:** (2026-08-24, File ▶ **Project** — a new, opt-in FAIR provenance record (HDF5), separate from GUI State: while a project is open, every completed Calibrate/Batch Integrate run (single-detector or per Hydra panel) automatically appends a self-contained record of its exact inputs, parameters, results, and software versions — raw multi-frame datasets are referenced by path + checksum rather than duplicated, so this stays small even for scans with thousands of frames. See §17 “File ▶ Project (FAIR provenance)”.)

**Previously:** (2026-08-24, Tab 4 — Batch Integrate: split into **Single detector** / **Hydra** modes behind a leftmost mode ribbon, the same pattern as the Calibrate/Data Viewer tabs’ own splits. Hydra mode integrates each of the 4 GE panels with its own independently fitted geometry — auto-populated from the Calibrate tab’s Hydra fits as each panel finishes, or loadable from a file per panel — using one shared Integration/Corrections/Monitor -normalisation/Output recipe, a **Sequential** or **Parallel** run-mode choice, independent per-panel masks, and per-panel Waterfall/Stacked -profile output. See §7 “Hydra mode (4-panel GE detector)”.)

**Previously:** (2026-08-24, Tab 2 — Calibrate: split into **Single detector** / **Hydra** modes behind a leftmost mode ribbon, the same pattern as the Data Viewer tab’s own split. Hydra mode fits each of the 4 GE panels’ own geometry from one calibrant

dataset using a shared pipeline/refine recipe, with independent per-panel Transforms/seed/results, a **Sequential** or **Parallel** run-mode choice, and a ← **Data Viewer** import of the Hydra Data Viewer page's loaded panels + fitted geometry. See §5 “Hydra mode (4-panel GE detector)”.)

**Previously:** (2026-08-24, Tab 0 — Data Viewer: the **Transforms** (Flip Y/Flip Z/Transpose[/Rotate]) checkboxes are now their own boxed **Transforms** card, directly below the Ring simulation card, instead of an inline “Transforms:” label + row inside it — fixes a large visual gap between the heading and the row; this is shared code, so it applies to both the single-detector tab's geometry card and every Hydra-tab panel card. Hydra mode: the **Projection** card moved from the left-side loader panel to the top of the middle panel, matching where the single-detector tab's own Projection card sits; and the radial-integration plot's pan/zoom is now bounded to the combined data range of the currently-visible curves (same margin formula as the single-detector tab's radial plot), instead of being free to scroll/zoom arbitrarily far from the data.)

**Previously:** (2026-08-24, Tab 0 — Data Viewer, Hydra mode: added a per-panel **Projection** card (Max/Sum/Average stack reduction, mirroring the single-detector tab) that feeds all of GE1-4 and the Composite view; added a per-panel-only **Rotate** field (clockwise, degrees) next to Flip Y/Flip Z/Transpose on GE1-4 — deliberately excluded from the Composite view; fixed a bug where Flip Y/Flip Z/Transpose didn't actually refresh the displayed image on GE1-4 panels; and the Material dialog's **Preset** dropdown now also fills in the **Name** field (still editable), on both the single-detector and Hydra tabs.)

**Previously:** (2026-08-24, Tab 0 — Data Viewer, Hydra mode: fixed a manual (Phase 6) review's 5 bugs —  $\lambda/\text{max } 2\theta/\text{pixel size}$  now shared across all 5 geometry cards; dark/bright/background correction added (sibling -aware, mirroring the single-detector tab); the Composite windmill assembly's panel rotation stays counterclockwise (confirmed correct against real `test_data/slide` data) and now also mirrors the whole canvas about its vertical axis, which was needed to put ge2/ge4 on their correct sides; a beam-centre edit now correctly updates the radial profile, not just the ring overlay; and the image viewer's `vmin%` auto-level now excludes exact -zero pixels app-wide, fixing a washed-out Composite view. The intensity -range exclude mask and Top-N brightest-pixel remain single-detector-only.)

**Previously:** (2026-08-23, Tab 0 — Data Viewer, Hydra mode: the radial plot now shows all 4 panels' profiles at once (fixed colors, independent per-panel calibration) plus a toggleable summed **Composite** curve, replacing the earlier single-curve placeholder. Dark/bright/mask corrections for Hydra frames are still a planned follow-up.)

**Previously:** (2026-08-23, Tab 0 — Data Viewer: the **Hydra** mode ribbon entry became functional — 4-panel GE detector loading, ge1-4/ composite image toolbar, independent per-panel beam-centre/ring calibration, and a geometry-based windmill composite. Single detector mode unchanged and still the default.)

**Previously:** (2026-08-23, Tab 0 — Data Viewer: added a leftmost **mode ribbon** (Single detector / Hydra) — Hydra mode was a placeholder at that point.)

**Previously:** (2026-08-23, All detector-image viewers — Data Viewer, Mask Builder, Calibrate — now draw pixel `(0, 0)` at the **bottom-left** corner instead of the top-left, matching MIDAS's convention that the on-screen image match the physical world view of the detector when looking downstream from the sample along the beam direction. This is a display-only rendering change (`pg.ImageView`'s own default `invertY()` is overridden); it is independent of the **Transforms** checkboxes and the `ImTransOpt` data transform, which are unchanged. The Data Viewer ROI tool's box annotations now label the bottom-left corner (was top-left).)

**Previously:** (2026-08-17, Tab 2 — Calibrate: the **Transforms: Flip Y / Flip Z / Transpose** checkboxes now actually do something — toggling one live-updates the image preview and Pick BC/Pick Ring clicks land in that same transformed space, matching Data Viewer/Mask Builder; the calibration run itself now applies the identical transform to the calibrant image and Dark right before the pipeline call, fixing a real bug where several pipeline modes (Four-stage, Bayesian, Joint-cake, panel-layout, partial distortion-coefficient selection) silently transformed the image internally while the seed beam centre and detector dimensions stayed untransformed; → **Send to Data Viewer** now carries the Transforms state too. Also: the Average frames card's **skip** (stride) control was removed — **start/ end (0=all)** remain.) (2026-08-17, Data Viewer, Mask Builder, and Calibrate tabs all gain matching **Transforms: Flip Y / Flip Z / Transpose** checkboxes applying MIDAS's `ImTransOpt` image transform to the raw detector frame before display/masking/calibration; the Calibrate tab's existing transform controls now round-trip through saved/loaded `paramtest.txt` and `calibration.json` files instead of being in-memory-only, and geometry hand-offs between the three tabs keep the Transforms state in sync.)

**Previously:** (2026-08-16, Tab 2 — Calibrate: two fixes. (1) Picking a strict subset of distortion coefficients (the “...” dialog) now genuinely restricts the LM fit to just those coefficients on every pipeline, including One-shot — which previously silently refined all 15 regardless of the selection — and the Results tab now lists only the coefficients actually selected for that run instead of always showing all 15 (saved `paramtest.txt/.json` exports are unaffected — they still carry every slot's real, possibly held-fixed, value). (2) Selecting/changing a Dark, Bright, or Background field now immediately updates the image preview to the corrected frame — it previously kept showing the raw, uncorrected image until the next full data load).

**Previously:** (2026-08-13, Data Viewer ROI tool usability follow-ups: fixed an OS-level minimize glitch where a popup would pop back open needing a second click; box ROI  $(x, y) / w = / h =$  annotations are now whole pixels, not fractional; line ROIs now show their length in pixels at the line's midpoint, and their "ROI N" label is anchored at the line's actual start point instead of the image's top-left corner; and the box/line popup plots plus the Intensity statistics histogram now cap zoom-out/pan at the current data range, matching the radial-integration plot) (2026-08-13, Data Viewer ROI tool: OS-level window minimize now tucks a popup into the ribbon (the old custom "—" button is gone); box ROIs show live  $(x, y) / w = / h =$  annotations on the image; all on-image ROI text gets a translucent background and a ~20% larger font; popup stats/crop images now respect the active bad-pixel mask (excluded from histograms/line stats, marked bright red on the crop); and the line-ROI drag preview is a true diagonal line instead of a box) (2026-08-12, Tab 1 — Mask Builder's **Dilation (px)** control switched from 4-connected to 8-neighbor dilation: N=1 now grows a bad pixel to the full surrounding 3x3 block, N=2 to the full 5x5 block, etc.) (2026-08-12, Tab 1 — Mask Builder gains a **5 · Post-processing** card with a **Dilation (px)** control that grows bad-pixel regions from the threshold/statistical/loaded mask by N pixels before combining with hand-drawn shapes) (2026-08-12, Tab 1 — Mask Builder: Stack browse menu gains a **Files (multi-select)...** entry so the temporal stack can be built from an explicit, hand-picked set of files instead of only a whole folder/glob) (2026-08-12, Tab 6 — PDF Analysis rebuilt for the full ROADMAP Stage 2-3 workflow: empty-cell/Paalman-Pings background subtraction, detector-efficiency correction, absolute normalization, differentiable multiple scattering, a fluorescence diagnostic, CIF-driven structure refinement, and  $\Delta$ -PDF significance testing, backed by a dedicated `test_data/test_pdf/ dataset`) (2026-08-10, Preferences ▶ Profile ships three bundled beamline device presets — **20-ID-D**, **20-ID-E**, **1-ID-E** — so the Data Viewer's Live Data PV dropdown can be switched to a beamline's detectors without hand-editing Preferences ▶ Devices) (2026-08-09, Cross-tab data sharing: every Data Loader panel's Data browse button, plus Mask Builder's Image/Stack browse buttons, gain an **Import from...** menu to pull a file/folder/buffer already loaded in another tab; **Use Buffer** gets a  button to save the buffer to an HDF5 file (`buffer/data`); Data Viewer's **Project stack** button turns green while a projection is displayed; ROI popups are now always-on-top and can be minimized to a ribbon on the image viewer's left edge) (2026-08-09, Data Viewer: new B-PILOT auto-start bridge — a local-socket server lets the separate B-PILOT plan-runner GUI trigger Live Data on a scan's detector with no clicks in MIDAS GUI; see the Live Data card section) (2026-08-03, Data Viewer/Calibrate/PDF: the clickable  $\lambda$  label's popup menu gains an **Energy (keV)** entry box that converts to wavelength on Enter, ahead of the existing K-edge foil menu; Data Viewer's Projection card drops the **Axis** field (always 0, i.e. across frames) and adds an **N frames** field capping how many frames after Skip frames are included, 0 = all remaining) (2026-08-01, Data Viewer: box-ROI popup's zoomed crop image is no longer fixed-size — it now resizes along with the popup window) (2026-08-01, Data Viewer: line ROI is now drawn as a single arrow shape — shaft and head recomputed together from the two endpoints on every drag — replacing the separate arrowhead overlay item that could rotate oddly as the endpoint moved) (2026-08-01, Data Viewer: image toolbar gains a Box/Line ROI tool — click-drag draws a shape, opening a small floating, freely-draggable stats popup next to it, color/label-matched to the shape; box popups show a linear/log intensity histogram plus a zoomed-in crop of the region, line popups show a flippable intensity-vs-distance profile; popups stay live-linked to their shape as it's dragged/resized but not to its screen position; multiple ROIs supported, session-only, removable via the popup's close button, a shape's right-click menu, or Clear ROIs (which also resets ROI numbering back to 1); Pick BC/Pick Ring's shared Clear button now also removes the Pick BC crosshair marker, not just Pick Ring's points) (2026-07-31, Data Viewer: pixel-size field now accepts a second decimal place, needed for near-field detectors' finer pixel pitches; radial-integration plot gains a "?" help button explaining the R-bin calculation; ring-width field widened 60->80px; Live Data card gets a "Use Buffer" last-N-frames ring buffer — yellow while filling, green once streaming pauses, at which point Projection and the rest of the stack analysis work on it like a loaded HDF5/folder stack; Load-calibration card renamed/moved below Ring simulation's Simulate button and now syncs ty/tz on load; Intensity range card's title bar is now the on/off checkbox; Mask rows — including the "Tab 1 mask" from the Mask Builder tab — get a checkbox to include/exclude without deleting, and the Tab 1 mask row always stays at the top of the list; Live Data "Use Buffer" N field capped at 100 frames to bound memory use; Mask status line turns amber and names any source dropped for a load failure or shape mismatch, instead of silently ignoring it; Live Data PV dropdown gains a built-in **Sim Detector** entry — a hardware-free fake PVA stream shaped like an Eiger2 500K with 0-60000 counts, for exercising Live Data without a real beamline connection, see `midas_gui/sim_detector.py`; image viewer colorbar/histogram now defaults its own zoom to the vmin%/vmax% percentile window instead of the full data range, converts a manual level/zoom window between Log and Linear scale on toggle, auto-resets to the percentile defaults whenever new data is loaded — except frame-to-frame updates within an active live stream, which keep a manual window fixed — and always reframes its own visible window tightly around the (converted) levels on a Log/Linear toggle, instead of carrying a manually zoomed/panned window through the nonlinear conversion, which could leave the sliders squeezed into a barely-visible sliver of the window; Live Data "Use Buffer" now carries over into **Start** instead of being silently reset — if buffering is already armed when Start is clicked, it re-arms with a fresh buffer for the new stream rather than turning back off; Ring simulation card's  $\lambda$ /Lsd/pixel size fields now accept any positive value (no more 1–5000  $\mu\text{m}$  pixel-size cap etc.) and display  $\lambda$  to 4 decimals, Lsd to 3, pixel size to 2 (needed for near-field detectors' sub- $\mu\text{m}$ -precision pixel pitches), BC\_y/BC\_z to 1, and ty/tz to 2; "Show rings" renamed **Rings** and moved onto the same row as **Labels** and a shrunk ring-thickness field; Exclude-out-of-range-pixels controls moved from their own left-panel card into the radial-integration plot's own toolbar (right end of the `X / Log Y / R bin / Auto / Integrate` row), with the `R bin` field narrowed and the `N bins | max=...` stats printout removed from that row to make space; Load calibration card renamed **Load/save calibration**; Exclude-range controls further refined — moved to sit pinned at the far right of the toolbar row, the `</>` bound fields widened (56px → 112px) and switched from float to integer spin boxes (no decimal

display), and the lower-bound comparison changed from `<=` to a strict `<` — a pixel exactly equal to the lower bound is no longer masked)

**Maintenance:** keep this document in sync with the code — whenever the workflow or a tab’s controls change, update the relevant section here in the same change.

**Tab status:** Tabs **0–4** (Data Viewer, Mask Builder, Calibrate, Calib. Refinement, Batch Integrate) are **verified** and ready to use. Tabs **5–8** (Corrections & Physics, PDF Analysis, Texture, Results & Export) are a **work in progress** and will be updated in the coming weeks.

---

## Table of Contents

1. [Overview and Architecture](#)
2. [Getting Started](#)
3. [Data Viewer](#)
4. [Mask Builder](#)
5. [Calibrate](#)
6. [Calibration Refinement](#)
7. [Batch Integrate](#)
8. [Corrections & Physics](#)
9. [PDF Analysis](#)
10. [Texture / Pole Figure](#)
11. [Pump Probe \(time-resolved / TR-XRD\)](#)
12. [Results & Export](#)
13. [Common UI Conventions](#)
14. [Packaging, Deployment & Diagnostics](#)
15. [Configuration & Defaults](#)
16. [File ▶ Save/Load GUI State](#)
17. [File ▶ Project \(FAIR provenance\)](#)

---

## 1. Overview and Architecture

The MIDAS GUI is a modular PyQt5 desktop application (up to 10 tabs) that exposes the scientific capability of the `midas_calibrate_v2` and `midas_integrate_v2` packages through a structured workflow. The intended order of use is:

```
Data Viewer (inspect) → Mask Builder → Calibrate → [Refine] → Batch Integrate
                                     → Corrections preview
                                     → PDF Analysis
                                     → Texture / Pole Figure
                                     → Pump Probe (TR-XRD)
                                     → Results & Export
```

**Modular tabs.** Data Viewer, Mask Builder, Calibrate and Batch Integrate are always shown; the remaining tabs (Calibration Refinement, Corrections, PDF Analysis, Texture, Pump Probe, Results & Export) are optional and can be shown/hidden from **Settings ▶ Preferences ▶ Tabs**. By default only **Calib. Refinement** and **Pump Probe** are shown; Corrections, PDF Analysis, Texture and Results & Export ship **hidden** — turn them on when you need them. The choice is saved per-user (`ui.visible_tabs`) and applies immediately — see §15. Hidden tabs are only removed from the tab bar; they stay constructed, so cross-tab wiring and state are preserved.

**Cross-tab shared state.** When Tab 2 (Calibrate) produces a result it is automatically propagated to all downstream tabs. When Tab 1 (Mask Builder) computes a mask it is sent to the consuming tabs. Tab 3 (Refinement), if applied, re-broadcasts the refined geometry. No manual copying is needed.

**Cross-tab data import (“Import from…”).** Every Data Loader panel’s **Data** browse button (Data Viewer, Calibrate, Calib. Refinement, Batch Integrate, Pump Probe), plus Mask Builder’s **Image** and **Stack** browse buttons, carry an **Import from...** submenu listing whatever’s currently loaded in every *other* tab — a file/folder path, or, if that tab’s Live Data **Use Buffer** ring buffer is frozen (green), a **Buffer (N frames)** entry. The menu is built fresh each time it’s opened, so it always reflects what’s loaded right now. Picking a path just loads it like a normal browse. Picking a **buffer**: - In a tab that keeps a live in-memory stack (Data Viewer, Calibrate, Refinement), the picking tab **delegates** to the source’s buffer directly — no copy is made, so it always reflects the source buffer’s current contents, and if the source buffer is later reset, the delegating tab clears itself and shows

“Source buffer was reset” rather than showing stale data. - In a tab that only ever reads from a file/HDF5 path (Batch Integrate, Pump Probe, and Mask Builder’s Stack field), the buffer is instead **snapshotted once** to a temporary HDF5 file (dataset `buffer/data`) and that path is loaded — later changes to the source buffer are not reflected until you re-pick the entry.

**All heavy computation runs off the GUI thread.** Every operation that touches a MIDAS package (calibration, integration, mask training, PDF transform, gain training, field averaging, drift fitting) runs in a background `QThread` worker; the GUI stays responsive and log lines stream in real time.

### Package layout.

```
midas_gui/
├─ app.py           MainWindow, dark theme, crash diagnostics, main()
├─ __main__.py      enables `python -m midas_gui`
├─ style.py         Dioplas-inspired QSS theme + layout helpers
├─ constants.py     calibrants, colormaps, dtype sentinels, default paths
├─ helpers.py       image IO, geometry parsing, spec building, no-scroll widgets
├─ widgets.py       ImageViewer / PickableImageViewer / ProfileViewer / FieldSelector / ...
├─ workers.py       all QThread workers + the integration core
├─ calib.py         calibration-pipeline dispatch
├─ dialogs.py       save dialogs
└─ tab_*.py         one module per tab
```

**Image orientation.** Every detector-image viewer (Data Viewer, Mask Builder, Calibrate) draws pixel `(0, 0)` at the **bottom-left** corner, matching MIDAS’s own convention: the on-screen image then matches the physical world view of the detector when looking downstream from the sample along the beam direction. This is a display-only convention, separate from the **Transforms** (Flip Y / Flip Z / Transpose) checkboxes found in Data Viewer, Mask Builder, and Calibrate — those still flip or transpose the underlying pixel *data* itself (persisted as MIDAS’s `ImTransOpt` in saved calibration files) to correct a detector’s raw readout orientation, independent of which corner the GUI renders as the origin.

**Layout pattern.** The four analysis tabs — **0 Data Viewer, 2 Calibrate, 3 Calib. Refinement, 4 Batch Integrate** — use a **three-panel layout**:

```
[ Data Loader | Parameters | Display / results ]
```

- **Data Loader (left).** A shared `DataLoaderPanel` for selecting the five inputs — **Data, Dark, Bright, Background, Mask** — each as a single file, a folder, or an HDF5 dataset (a container dropdown appears for HDF5). Frame controls live here too (Tab 0 navigator, Tab 2/3 frame index, Tab 4 frame range + stride).
- **Parameters (middle).** The tab’s analysis controls.
- **Display / results (right).** Image viewer, plots, and/or a log panel.

The three panels are separated by **draggable splitter handles** — drag the `Data Loader | Parameters` and `Parameters | Display` boundaries to rebalance widths to taste. Each panel has a minimum width; drag past it and the panel shows a scrollbar rather than clipping its controls.

The remaining tabs (1, 5–8) keep the classic two-panel layout.

**Corrections & mask (all four analysis tabs).** Dark is averaged then subtracted; Bright is averaged then flat-field **divided** or **subtracted** (your choice); Background is averaged then subtracted; every **checked** Mask source (files/folders plus the auto-added “Tab 1 mask”, populated from the Mask Builder tab) is **unioned** into one composite mask that zeroes/ignores those pixels. Correction order: `(img - dark) → bright → - background → clip ≥ 0`.

Each Mask row has its own **checkbox** to include/exclude it from the union without deleting it (the **X** button still removes the row entirely). The “Tab 1 mask” row is always kept at the **top** of the list when other mask sources are present.

If a checked mask source fails to load, or its shape doesn’t match the other sources already unioned, it’s dropped from the composite and the status line under the mask list turns amber and names the source and the reason — it is never silently ignored.

---

## 2. Getting Started

`midas-gui` is **not on PyPI** — install it from source with conda.

```
git clone https://github.com/d-beniwal/MIDAS_GUI.git
cd MIDAS_GUI
conda env create -f environment.yml # creates the 'midas-gui' env and installs the GUI
conda activate midas-gui
midas-gui # launch (equivalently: python -m midas_gui)
```

The GUI drives the MIDAS analysis backends (`midas-calibrate-v2`, `midas-integrate-v2`, `midas-calibrate`, `midas-hkls`, `midas-distortion`), which are also not on PyPI — point the `pip:` section of `environment.yml` at your MIDAS source before creating the environment (see the comments in that file).

## Test data

Synthetic test data lives in the repo-root `test_data/` folder (git-ignored): - `calibrant_ceria.tif / .h5` — CeO<sub>2</sub> calibrant (Eiger2 500K, 1028×512, 75  $\mu$ m pixels) - `nickel_tifs/` — 10-frame Ni scan, lattice expanding +0.1 %/frame - `nickel_stack.h5` — the same scan as one HDF5 stack (`exchange/data`)

Default geometry:  $\lambda = 0.39 \text{ \AA}$ , Lsd = 121 mm, pixel = 75  $\mu$ m, BC = (10, 10) px. Every tab's default file paths point at this data and **auto-load on startup when present**, so the app is usable immediately after checkout.

## 3. Tab 0 — Data Viewer

**Purpose:** inspect detector frames and do a quick radial integration — geometry-free (circle binning) by default, or a full tilt/distortion-aware integration when a calibration file is loaded. Produces no shared state for other tabs.

### Mode ribbon (leftmost strip)

A narrow vertical strip at the very left edge of the tab switches between **Single detector** (the view described below — unchanged) and **Hydra** (the 1-ID-E 4-panel GE detector view). The two modes are independent: switching does not share data or geometry between them. **The Hydra option only appears when the active profile (top-left header dropdown — see §15 “Profiles”) is 1-ID-E**, the only beamline with that detector; on any other profile the ribbon shows Single detector only, and switching away from 1-ID-E while Hydra mode is active falls back to Single detector automatically. Same rule applies to Calibrate's (§5) and Batch Integrate's (§7) identical mode ribbons.

### Hydra mode (4-panel GE detector)

**Purpose:** inspect and calibrate the 1-ID-E Hydra detector — 4 separate GE panels arranged in a “windmill” layout around a shared beam axis, each with its own independent beam-centre/tilt calibration, composited into one registered image for a full-coverage view.

- **Hydra data (left panel):** point the path field at **any one** of the 4 GE panel files (a `.geN.h5 / .tif` or a `geN/` folder — matching this beamline's own naming convention) and the other 3 panels are found automatically. Small `ge1 ge2 ge3 ge4` status labels turn green as each panel is located (grayed out if not found — the view still works with as few as 2 panels present). A frame slider/spinbox navigates a shared frame index across all panels (they're synchronized frames of the same scan).
- **Dark / Bright / Background (left panel, below Hydra data):** same correction math as the single-detector tab (dark subtraction, bright flat-field divide/subtract, background subtraction). Point each field at **any one** panel's dark/bright/background file and the other panels' matching files are found automatically, the same way the main Hydra data path works — no need to pick all 4 by hand. Each field computes and applies independently per panel.
- **Projection (top of the middle panel):** same **Max / Sum / Average** stack-reduction as the single-detector tab's own Projection card (and in the same place — top of the middle panel, above the geometry cards), but computed **per panel** — clicking **Project stack** reduces every currently-found panel's own frame stack (honoring Skip frames/N frames and whatever Dark/Bright/Background correction is currently set) and shows the result in place of the current frame for **all of GE1-4 and the Composite view** (the Composite is rebuilt from the projected frames, not frame 0). **Back to frames** — or moving the frame slider/spinbox — returns to normal per-frame navigation.
- **Image toolbar:** five buttons — **GE1 / GE2 / GE3 / GE4 / Composite** — select what's shown in the image viewer below. GE1-4 show that panel's own raw frame; **Composite** shows all currently-available panels remapped (via each panel's own beam-centre + tilt) into one shared, registered canvas — a geometry-based “windmill” composite, not a raw mosaic.
- **Per-panel geometry card (middle panel, below the Projection card):** switching the image-toolbar button swaps which panel's **Ring simulation + Transforms + Load/save calibration** cards are shown — identical in every way to the single-detector tab's own cards (materials list, beam-centre pick/ring-fit, tilt fields, calibration file load/save), just bound to that one panel. Loading a calibration file (or picking/editing a beam centre) on a GE1-4 card takes effect immediately: it's used



for that panel's own ring overlay/radial integration, and — since the Composite view depends on every panel's geometry — the composite is automatically rebuilt the next time it's shown. Panels with no calibration loaded yet use a bundled default 1-ID-E-style geometry (an example windmill layout, not your instrument's real calibration — load a real per-panel file to override). The **Composite**'s own geometry card is separate from the 4 panels': its beam centre is automatically seeded at the composite canvas's own centre the first time a given canvas size is built, so ring simulation/radial integration on the Composite view work immediately with no extra setup (though it can still be hand-edited like any other card).  **$\lambda$  (wavelength), max 2 $\theta$ , and pixel size are shared across all 5 cards** — editing any one of them on GE1-4 or Composite applies it to the other 4 immediately (same X-ray beam and GE detector model, so these three are always physically identical); beam centre, Lsd, and tilt remain independent per panel. GE1-4 (not the Composite) also have a **Rotate** field on the same row as **Flip Y / Flip Z / Transpose** in that panel's own **Transforms** card: a clockwise rotation (degrees, default 0) applied only to that one panel's own raw display/radial-integration image — it is deliberately **not** applied to the Composite view, which keeps building from each panel's un-rotated frame and its own independent beam-centre/tilt geometry.

- **Radial integration plot (bottom right)**: shows **all 4 panels' own azimuthal profiles at once** — GE1-4 in fixed colors, computed independently from each panel's own beam centre/geometry — plus a toggleable **Composite** curve (white, dashed). Checkboxes above the plot show/hide each curve individually. The R-bin/Auto/Integrate controls are shared across all 5 views (one setting, not a separate control per panel); **Integrate** recomputes every curve immediately regardless of Auto. The X-axis unit selector (R/2 $\theta$ /Q) applies to all curves at once. The **Composite** curve is the sum of the 4 panels' own profiles — each resampled onto a shared 2 $\theta$  axis first, then added together — **not** a radial integration of the composited image itself (which would double-count any panel overlap and mix registration error into the profile). **Pan and zoom are bounded** to the combined X/Y extent of every currently-visible curve (same margin formula as the single-detector tab's own radial plot — see **Pan and zoom are always bounded to the current profile's extent** below), recomputed on every curve refresh (X-axis unit switch, checkbox toggle, or new data) so scrolling/zooming can't wander off into empty space. This plot doesn't yet have the single-detector plot's Auto/Manual toggle button pair.
- **Not yet supported in Hydra mode** (present in Single detector mode): the intensity-range exclude mask, and Top-N brightest-pixel.

Data Loader panel (left)

Data, Dark, Bright, Background and Mask are all selected in the shared **Data Loader panel** (see §1): each accepts a file / folder / HDF5-dataset. The **Data** card here holds the **frame navigator** (slider, spin box, ◀/▶); **zoom/pan is preserved** as you step through frames (the view only auto-frames on a fresh load). Dark/bright/background and the composite mask are applied to the displayed image and to the radial integration. **Pan and zoom are bounded to the image** (roughly half an image-width of margin on each side) so scrolling/dragging cannot wander off into empty space or zoom out indefinitely; the bottom-left **A** (auto-range) button re-fits the image without leaving zoom “stuck” tracking the mouse. On the **1-ID-E / 20-ID-D / 20-ID-E** profiles, loading a Data file also auto-detects **pixel size** (from a .ge1 — .ge5 / .vrx / .pxrd filename tag) and, for an HDF5 file, **wavelength** (from its recorded beam energy) into the Ring simulation card's  $\lambda$ /pixel fields below — whatever isn't detected is left unchanged.

Projection card

Collapse a stack to one image: **Max** (hot-pixel hunting), **Sum** (long-exposure equivalent), or **Average** (noise reduction), always across the stack of frames. **Skip frames** (default 1) ignores that many leading frames before projecting (e.g. 1 drops the first frame, 4 drops the first four) — useful when the opening frames are detector warm-up / shutter-transient exposures. **N frames** caps how many frames (after Skip frames) are included in the projection; **0** (default) uses every remaining frame in the stack. While a projection is being displayed, **Project stack** stays highlighted **green** so it's obvious the image on screen is a projection rather than a live frame; **Back to frames** (or loading new data) reverts it to its normal styling.

Exclude-out-of-range-pixels controls (radial-plot toolbar)

These controls used to be their own left-panel card; they now live at the **far right** of the **radial integration plot's toolbar** (see below), past the **X**, **Log Y**, **R bin**, **Auto**, **Integrate** controls, so there's no separate card here anymore.

| Field                    | Description                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exclude range (checkbox) | When on, pixels < min or > max are drawn as a red overlay and excluded from the radial integration (removes gaps / hot / overflow).                               |
| < / >                    | Lower / upper bounds, entered as <b>whole-pixel-count integers</b> (no decimals). On load, the upper bound auto-fills to <b>max(99.99th percentile, 100000)</b> . |

## Ring simulation card

Overlays simulated Debye-Scherrer rings to check geometry. Supports **multiple materials at once** — e.g. a sample phase overlaid on a calibrant — each with its own lattice, visibility, and ring color. - **Material rows**: one row per material, each with a **checkbox** (show/hide that material's rings on both the image and the radial-integration plot), a **color swatch** button (click to open a color picker; the chosen color drives that material's ring lines and hkl labels on both plots), and the **material name** as a clickable (underlined) button. A **X** button deletes the row — disabled when only one material remains, since at least one row is always kept. + **Add material** appends a new row (default name `Material N`, generic cubic lattice, next color from a 10-color palette cycled by row order). A single default **Ni (FCC)** row is present at startup, matching the pre-multi-material behavior. - Clicking a material's name opens a **Material dialog**: an editable **Name** field, the **Preset** dropdown (`CeO2`, `LaB6`, `Si`, `Al2O3`, `Cu`, `Ni`, `FCC-γFe`, `BCC-αFe`, `Au`, `Ag`, `Pt`, `W`, `Ti`, or **Custom**), lattice **a**, **b**, **c** (3 decimals) on one row and **a**, **β**, **γ** (2 decimals) on the next, plus **SG #**, and a **Cubic (a=b=c, α=β=γ=90°)** checkbox that lets you enter only **a** for cubic crystals (b, c mirror a; angles fixed at 90°). Lattice fields are editable only for Custom. Picking a named preset also fills the **Name** field with that preset's name (still freely editable afterward — e.g. rename it without losing the lattice values just applied). OK applies the name/lattice/preset back to that material's row (renaming here updates the row's displayed name); Cancel discards edits. - Geometry ( $\lambda$ , max 2 $\theta$ , Lsd, pixel size, beam centre) is shared across all materials — only the lattice/space-group/color/visibility are per-material. Beam centre (auto = image centre, or manual BC<sub>y</sub>/BC<sub>z</sub>). The  $\lambda$  label is clickable (underlined) — click it to open a menu with an **Energy (keV)** entry box at the top (type a photon energy and press Enter, or click  $\leftarrow$ , to convert it to wavelength via  $\lambda = 12.398420 / E$ ) followed by a menu of common K-edge foils (Pr, Sm, Yb, Lu, Hf, Ta, W, Re, Pt, Au, Pb, Bi) that set  $\lambda$  directly to that element's K absorption-edge wavelength. The same clickable- $\lambda$  menu is on the Calibrate and PDF tabs. The **px** label is likewise clickable — a menu of common detectors (GE 200  $\mu\text{m}$ , Varex 150  $\mu\text{m}$ , Pilatus 172  $\mu\text{m}$ , Eiger 75  $\mu\text{m}$ ) sets the pixel size (also on the Calibrate tab, where it sets both pxY and pxZ). - Every field in this card steps by a fixed amount per up/down-arrow click ( $\lambda$  0.01 Å, max 2 $\theta$  1°, Lsd 1 mm, pixel 0.1  $\mu\text{m}$ , BC<sub>y</sub>/BC<sub>z</sub> 1 px, ty/tz 0.1°) — customizable from **Settings** ▶ **Preferences** ▶ **Data Viewer**. - **ty / tz** (detector tilt about the Y/Z axes, degrees) sit right below BC<sub>y</sub>/BC<sub>z</sub> — when non-zero, the simulated rings are forward-projected through the tilt geometry instead of drawn as plain circles, so the overlay shows the same non-circular ring shape a tilted detector actually produces. Leaving both at 0° reproduces the previous plain-circle rendering exactly. (Rotation about the beam axis, **tx**, leaves a full ring's shape unchanged, so it isn't exposed here.) - **Rings / Labels** toggles sit on one row together with a compact **thickness** spin box (0.5–10 px) that sets the line width of the simulated-ring overlay on the image; redraws immediately as you change it. - **Simulate rings** is a **live toggle** (not a one-shot click) — while on, the button turns **green** as a visual cue, and the overlay + hkl table recompute automatically whenever material, lattice constants, space group, or geometry ( $\lambda$ /Lsd/px/max 2 $\theta$ ) change. Beam-centre and ty/tz edits still just reposition the existing rings (their 2 $\theta$  values don't depend on BC or tilt). Rings are drawn out to the full **max 2 $\theta$**  you set, regardless of how far that places them from the beam centre (rings landing off-canvas simply aren't visible — they aren't silently dropped at some fixed pixel-radius cutoff). -  $\rightarrow$  **Send geometry to Calibrate** copies  $\lambda$ , pixel size, Lsd and beam centre into the Calibrate tab's detector + seed fields (the Calibrate tab has a matching  $\leftarrow$  **Data Viewer** button that pulls the same values).

## Transforms card

A small standalone card sitting directly below the Ring simulation card, titled **Transforms** (previously an inline “Transforms:” label + row buried inside the Ring simulation card, with the label and row visually far apart — now its own boxed section so the heading sits directly above the checkboxes it labels). **Flip Y / Flip Z / Transpose** apply MIDAS's `ImTransOpt` image transform to the raw detector frame — before display, ring overlay, and radial integration — in that fixed order (flips, then transpose). Use this when the detector's raw pixel orientation doesn't match the geometry model (e.g. the beam centre would otherwise land on the wrong side of the image). Toggling refreshes the current frame (or the active projection) immediately. The same three checkboxes appear on the Mask Builder and Calibrate tabs and stay in sync whenever geometry is pushed/pulled between tabs; saved/loaded calibration files (`.json` / `.txt`) round-trip the codes as MIDAS's repeatable `ImTransOpt <code>` paramstest key (1=Flip Y, 2=Flip Z, 3=Transpose). In Hydra mode, GE1-4's Transforms cards also carry the per-panel-only **Rotate** field (see Hydra mode below); this same Transforms card is shared code between the single-detector and Hydra tabs.

## Load/save calibration card (optional)

Sits directly below the Transforms card. Load geometry from a **calibration** `.json`, a **MIDAS** `paramstest.txt`, or a **pyFAI** `.poni` (auto-detected). It fills **BC**, **Lsd**, **pixel size**, **wavelength**, the **Ring-simulation card's** **ty/tz** tilt fields, and the **Transforms checkboxes** (from any `ImTransOpt` lines in the file), unchecks “Beam centre = image centre”, and refreshes the overlay and radial plot.

When a calibration file carries the **full geometry (tilts + distortion)**, the radial integration switches from simple concentric-circle binning to a **proper MIDAS-engine integration** that maps every pixel through the calibrated tilts and distortion (the same core as Batch Integrate) — so ring positions/intensities are geometry-correct, not just distance-from-beam-centre. The card's status line reports which mode is active. The binning geometry is built once and reused across frames (a fast **hard** kernel keeps the preview responsive); without a full-geometry file, the fast circle binning is used. If no calibration file is loaded but the



Ring-simulation card's **ty/tz** tilt fields are non-zero, the radial integration still runs the tilt-aware MIDAS engine using a geometry built live from those fields (BC, Lsd, pixel size, wavelength) — so dialing in tilts here without a calibration file already produces a tilt-corrected profile, not just a tilt-shaped ring overlay.

**Save JSON / Save params (.txt) / Save PONI** (below the loader) export whatever geometry is currently in effect — the loaded calibration file if any, otherwise the one synthesized from the Ring-simulation widgets above — as a calibration file you can reload here, on the Calibrate tab, or in Batch Integrate. **Save params (.txt)** writes a standalone MIDAS `paramtest.txt`; **Save PONI** writes a pyFAI `.poni` (note: PONI's Rot1–3 convention cannot represent MIDAS's tx/ty/tz tilts, so tilts are **not** included in a `.poni` export — use JSON or `.txt` to keep them). Clicking any of the three before an image is loaded warns instead of writing a file (there is no detector size to write).

### Live Data card (*experimental*)

Sits **above the Data card**, collapsed by default behind its own title-bar checkbox — check it to reveal the live-PV controls, uncheck to hide them again (unchecking also stops an active stream, so a hidden card can never be left silently connected). Subscribes directly to an EPICS PVA detector-image PV (NTNDArray) and renders each frame **inline in the Data Viewer's own image pane** — the same view used for files/folders/HDF5 stacks. Because it feeds the normal frame pipeline, all existing Data Viewer analysis keeps working live: dark/bright/background/mask corrections, the intensity-range mask, beam-centre picking, and the radial integration plot all update as new frames arrive. - **Dependency:** `pvapy` (the EPICS pvAccess client library) is a required, pinned dependency (`pyproject.toml` / `environment.yml`), installed automatically with the rest of the GUI's stack — no separate extra to install. If somehow missing from the active environment, **Start** shows an install hint instead of failing outright. - **Live PV** field is an editable dropdown: pick a known device by name (the list comes from **Preferences ▶ Devices**, see below) to fill in its full PV automatically, or type any other PV by hand (placeholder shows an example, `2θIDFF:Pva1:Image`). **Start / Stop** buttons and a status line (stopped / waiting for PV / connected / streaming with frame id / error). GUI updates are throttled to the tab's existing ~16 fps debounce, so a fast PV update rate doesn't overwhelm the interface. - **Sim Detector** is a built-in dropdown entry (PV `midasSim:Pva1:Image`) for exercising Live Data with **no beamline hardware**: picking it and clicking **Start** lazily launches an in-process fake PVA server (`midas_gui/sim_detector.py`) that streams random frames shaped like a DECTRIS Eiger2 500K (1030×514 px) with counts in `[0, 60000]`, at 5 Hz by default — real `PvaLiveSource` code path, so it's indistinguishable from a real detector's stream. The rate, frame size and intensity range are constructor parameters in that file for anyone who wants a different fake stream. The simulator keeps running (harmless background thread) across repeated Start/Stop until the app closes, at which point it's stopped automatically. - **Use Buffer** captures the last **N** live frames (N field next to it, 2–100 — capped to bound memory use for large-format detectors) into an in-memory ring buffer, so Projection and every other stack-based analysis become available on live data. Click it to arm: it turns **yellow** and its label counts up (`Buffering... (7/20)`) while frames keep streaming in — the live single-frame view is unaffected during this. When no new frame arrives for ~2 s (streaming paused, or **Stop** clicked), it turns **green** (`Buffer Ready (20)`) and the buffered frames become a normal navigable stack: the frame slider/spin/prev-next enable, and **Project stack** runs max/sum/average over the buffered frames exactly as it would for a loaded HDF5 file or folder. If new frames resume arriving, it flips back to yellow and keeps rolling (oldest frame dropped once past N). Click it again to turn buffering off and discard the buffer. If **Use Buffer** is already armed (yellow or green) when **Start** is clicked, buffering carries over into the new stream — it re-arms with a fresh empty buffer rather than turning off, so you don't need to click **Use Buffer** again after Start. Loading static data always clears any existing buffer. A small  button next to **Use Buffer** enables once the buffer is frozen (green) — click it to save the buffered frames to an HDF5 file you choose, written as a (N,H,W) dataset named `buffer/data`. The same buffer is also what **Import from...** offers to other tabs — see **Cross-tab data import** in §1. - **Stopping live streaming does not auto-reload the previous file/folder/HDF5 source** — the Data card below gets a small  **Reload** button next to its path field for exactly this: click it after Stop to restore the static data that was loaded before streaming started. - **Ring overlay is static per-frame while streaming**: rings drawn by “Simulate rings” (Ring simulation card) keep rendering on top of each new live frame at their already-computed geometry — arrival of a new frame does **not** by itself trigger a recompute (ring radii don't depend on frame data). With Simulate rings' live toggle on, changing a material/lattice/geometry field still recomputes and redraws immediately. **Stop** leaves the last received frame on screen. Closing midas-gui also stops any running stream. - **Colormap/level changes persist across live frames**: dragging the histogram's level range or its own zoom (or the cmap dropdown) is remembered and reapplied to every new incoming live frame instead of being reset to the vmin%/vmax% percentile defaults. Editing vmin%/vmax%, toggling Log/Linear, or loading a new file/frame/dataset switches back to auto-levels — see §13 for the general rule shared by every image viewer in the app. - **B-PILOT auto-start bridge**: on launch, MIDAS GUI opens a local-socket server (`midas_gui/bridge_server.py`) that lets **B-PILOT** (a separate Bluesky plan-runner GUI) trigger Live Data with no clicks here — when B-PILOT dispatches a scan on a known detector, it sends a `{"type": "live_pv", "prefix": "...}"` message; MIDAS GUI resolves the prefix against **Preferences ▶ Devices** and, if it matches, checks/expands the Live Data card, fills in the PV and clicks **Start** automatically (switching PVs first if a different stream is already running). No-op — and silent in the log only — if B-PILOT never connects or the prefix isn't a known device.

### Beam-centre picking (on the image)

The image viewer has **Pick BC** (single click sets the beam centre) and **Pick Ring** (click  $\geq 3$  points on a ring; a circle fit estimates the beam centre). Either updates BC\_y/BC\_z and re-runs the overlay + radial integration. Pick Ring's picked points and fitted circle are drawn in **blue**, distinct from the **amber** Simulate-rings overlay so the two aren't confused when both are visible on the same image. **Clear** removes all of it at once — Pick Ring's points/fit **and** the Pick BC crosshair marker, regardless of which tool left it on screen.

## Top-N brightest pixels (image toolbar)

A **Top-N pixels** toggle button (with an **N** spin box) sits on the image toolbar. When on, the **N highest-intensity pixels** of the current frame are marked with a crosshair inside a translucent circle (cyan) centred on each pixel, and the Intensity statistics panel switches to show the statistics of just those N pixels ("Top N pixels"). It follows frame changes while active. Clicking the button again removes the markers and restores the normal statistics. Useful for quickly locating saturation / hot pixels.

An **I >** checkbox next to the N spin box enables an optional intensity floor (a field to its right, editable once the checkbox is on): when set, pixels at or below that value are excluded from ranking entirely — never marked, never counted toward N — so a low-signal frame can't be forced to mark N pixels that aren't actually meaningful. If fewer than N pixels clear the threshold, fewer than N markers are drawn (down to none).

## Region-of-interest (ROI) tool (image toolbar)

A **ROI: Box / Line** row sits on the image toolbar, alongside a **Clear ROIs** button. Click Box or Line to arm it, then click-drag on the image to draw the shape (a drag shorter than a few pixels is treated as a cancelled attempt — the mode stays armed so you can retry). Arming a shape mode automatically disarms Pick BC/Pick Ring and vice versa, since they all read the same click-drag. Drawing a shape opens a small **floating stats popup** next to it and un-arms the button (one-shot, like Pick BC). While dragging out a **line** ROI, the live preview is a true diagonal line (previously it looked like a rectangle until the mouse was released); a **box** ROI's drag preview is unchanged.

Each ROI gets its own color (cycled from a fixed palette) shared by the on-image shape, its on-image label, and its popup's title/label field, so multiple simultaneous ROIs stay easy to tell apart. The label is editable — typing a new name in the popup updates the on-image label to match. New ROIs are numbered "ROI 1", "ROI 2", ... in creation order; **Clear ROIs** resets that counter, so the next ROI drawn after a clear starts back at 1. All on-image ROI text (the "ROI N" label, and a box's corner/width/height annotations below) is drawn with a translucent dark background so it stays legible over bright image content, in a font ~20% larger than the app default.

A **box** ROI additionally shows its pixel geometry directly on the image, in the ROI's color, updating live while it's dragged or resized: the bottom-left corner's (x, y) coordinate (rounded to the nearest whole pixel) next to that corner, w = ... below the bottom edge, and h = ... to the right of the right edge (also rounded to whole pixels — no fractional-pixel values are shown). A **line** ROI shows its length in pixels (NN px, rounded) at the line's midpoint. Every ROI's editable "ROI N" label is anchored at the shape's actual position — for a line, that's its current start point (the end where distance = 0 in the popup's profile, swapping ends when **Flip direction** is used), not a fixed image corner.

- **Box** popups show a linear/log intensity histogram of the pixels inside the box, plus a zoomed-in crop of the boxed region rendered with the tab's current colormap. The crop image is not fixed-size — dragging the popup window's edge to resize it grows or shrinks the crop image along with the histogram. When a bad-pixel mask is active (composite mask file or the intensity-exclude controls), masked pixels are excluded from the histogram/stats and are marked in translucent bright red on the crop image, matching the main viewer's bad-pixel overlay convention.
- **Line** popups show a live **intensity-vs-distance profile** along the line (distance measured from one endpoint), plus N/min/max/mean of the sampled values. The line itself is drawn as a single arrow (shaft + head in one shape, recomputed from the two endpoints on every drag) pointing toward the end where distance = 0 → increasing runs; a **Flip direction** button in the popup reverses it. Masked pixels along the line are excluded from min/max/mean (shown as gaps in the profile); a line drawn entirely over masked pixels shows "(all pixels masked)" instead of the stats.

Both the box's histogram and the line's intensity-vs-distance profile cap how far you can zoom or pan out — the same range-limiting used on the tab's radial-integration plot — so scrolling/dragging the plot can't lose the data in an empty view; the limit is recomputed from the current data range each time the popup refreshes.

Dragging or resizing a shape (or right-click → drag a handle) recomputes its popup immediately, and every popup also refreshes automatically on frame navigation, new live-streamed frames, and dark/bright/background/mask correction changes — the same as the rest of the tab's live readouts.

A popup's on-screen **position is independent of the shape's position**: it opens next to its shape once, at creation time (placed on whichever monitor the shape is on, useful for multi-detector/multi-screen setups), and after that it's a normal free-floating window you can drag anywhere — moving it never repositions again on its own, only its contents stay live.

Closing a popup (its window close button) removes its shape from the image; right-clicking a shape and choosing **Remove ROI** closes its popup too. **Clear ROIs** removes every ROI and popup at once. ROIs are session-only — they are not saved with the rest of the tab's state.

**Always on top, minimize to ribbon.** ROI popups stay **above every other window** (not just above the main midas-gui window) so clicking elsewhere in the GUI, or in another application, never buries one behind something else. Minimizing a popup with the window's own **OS-level minimize button** (the title bar/traffic-light control) doesn't send it to the dock/taskbar as a normal window minimize would — instead the popup hides and a small colored square button, color-matched to its ROI, appears on a narrow **ribbon** along the left edge of the image viewer. Clicking a ribbon button restores that popup (shows it, raises it, gives it focus) and removes the ribbon entry. A minimized ROI's shape stays on the image and its stats keep updating live in the background exactly as if the popup were open — minimizing only hides the window. Removing a minimized ROI (right-click → **Remove ROI**, or **Clear ROIs**) also removes its ribbon entry.

### Intensity statistics (left panel, bottom)

At the bottom of the Data-Loader panel, in a **draggable pane** below the loader cards — grab the splitter handle above the panel to make the statistics area taller or shorter. It holds a **histogram** of the intensity distribution (full range, log-y toggle) with a **textbox** beneath it reporting N (pixel count) and the **p70 / p90 / p99 / p99.9 / p99.99** percentiles — each with the **number of pixels above** that value. The histogram's lower-left corner is fixed at  $x = 0, y = -2$  and both axes rescale to  $(0, x_{\max}) / (-2, y_{\max})$  on every refresh (frame change, scope change, projection, new data); zooming/panning is capped at that same range (as with the ROI popup plots and the radial-integration plot), so it can't be scrolled out until the distribution is lost in empty space. It reflects the **corrected** image (dark / bright / background) with masked pixels (file masks + the intensity-range mask) excluded, and updates live as any of those change. A scope selector switches between the **current frame** (per the slider) and **All frames** (combined over the whole stack/folder). When a **Projection** is active the panel shows the projected image's statistics (the scope selector is disabled); when **Top-N pixels** is active it shows those pixels' statistics.

A small “**A**”/“**M**” button pair sits in the histogram's bottom-left corner (replacing pyqtgraph's native auto-range corner button), same control as on the radial-integration plot below — see **Manual axis limits (A/M toggle)** under that section for the full behavior (native right-click “Manual” min/max fields, persistence across live updates, relick-to-reset). In **Manual**, the histogram holds those limits instead of auto-rescaling to  $(0, x_{\max}) / (-2, y_{\max})$  on every refresh (new frame, scope change, Top-N toggle).

### Radial integration plot (bottom-right)

Below the image is a live **azimuthal average about the beam centre** ( $R_{\text{bin}}$ , **Integrate**, and an **Auto** toggle that recomputes on frame/BC/mask change) — geometry-free circle binning by default, or the tilt/distortion-aware MIDAS engine when a full geometry is in effect (a loaded calibration file, or non-zero ty/tz in the Ring-simulation card — see the Load/save calibration card above). Peak/ring markers overlaid on the plot use the ring's true  $2\theta$ , so they line up with the profile in either mode. **Clicking a radius on the plot draws the matching ring (magenta) on the image.** Axis units switch between  $R$  (px) /  $2\theta / Q$ ; the **X-axis lower bound defaults to 0**. A small circular “?” button next to the **Radial** control opens a message box explaining how the profile is computed (full-geometry  $(\eta, R)$  binning vs. the circle-binning fallback). The **Exclude range** checkbox and its  $< / >$  integer bounds (see previous section) sit pinned to the far right end of this same toolbar row — the  $R_{\text{bin}}$  field was narrowed and the  $N_{\text{bins}} | \text{max}=\dots$  stats printout that used to occupy that space was removed to fit them.

**The default view always auto-fits the current profile's X/Y extent** — it never gets stuck showing a stale or unrelated range from an earlier profile. If you manually zoom or pan (drag, wheel-zoom, box-zoom, or the axis context menu), that exact view is **preserved across parameter changes** (new frame, changed BC/tilt/Lsd, etc.) instead of snapping back to full range every time the profile updates — the curve redraws in place under your current zoom. A manual zoom is only cleared when it would no longer make sense: switching the X-axis unit ( $R/2\theta/Q$ ) drops the remembered X range, and toggling **Log Y** drops the remembered Y range (both because the old numbers belong to a different scale). **Pan and zoom are always bounded to the current profile's extent** (a margin around its X/Y range) so you cannot drag or scroll off into empty space. The splitter handle above this panel (between it and the image view) is wider than a default Qt splitter, to make it easier to grab.

### Manual axis limits (A/M toggle)

A small “**A**”/“**M**” button pair sits in the plot's bottom-left corner, in the spot pyqtgraph's native auto-range button normally occupies (that native button is hidden in favor of this pair). **A** (default) is the auto-fit behavior described above. **M** switches to **Manual**, which holds exactly the limits set via each axis's own **native right-click menu** — right-click the plot, open **X axis** or **Y axis**, pick **Manual**, and type a min/max (this is stock pyqtgraph, not a MIDAS-specific control). Once **M** is active, the plot's axes show **exactly** those typed values (e.g. entering  $0$  shows  $0$ , not a padded/rounded value) and hold them through every live-acquisition redraw — Manual mode stops the plot from re-fitting or re-clamping its range on each new frame, and editing the min/max fields again takes effect immediately, live acquisition or not. Switching to **A** does not discard the typed values —

clicking **M** again restores exactly what was last entered. **Reclicking the already-active button resets the view to that mode's default**: **A** forces an immediate re-fit to the current profile (discarding any manual pan/zoom drift), and **M** snaps the view back to the held manual limits (discarding any drift from panning around while still in Manual).

---

## 4. Tab 1 — Mask Builder

**Purpose:** identify and exclude bad pixels. The final mask is a uint8 array (1 = bad) broadcast to Tabs 2, 3, 4, 5. Browsing an image or a mask loads it immediately. Pointing the **Image** field at an `.h5` file reveals a **Dataset** dropdown (auto-populated with the file's datasets and shapes, preferring a  $\geq 2$ -D dataset) — the same pattern used by the Stack field below it and by the Data Loader panels elsewhere in the app. The **Image** field's browse button also has an **Import from...** submenu of whatever file/folder is currently loaded in another tab (see **Cross-tab data import** in §1); the **Stack** field's browse menu additionally lists any other tab's frozen (green) Live Data buffer as a **Buffer (N frames)** entry — picking one snapshots that buffer to a temporary HDF5 file and feeds it into the auto-mask stack exactly like a browsed folder/file would. The Stack browse menu also has a **Files (multi-select)...** entry that opens a multi-select file dialog so the temporal stack can be built from an explicit, hand-picked set of files (e.g. frames scattered across a directory or spread across multiple directories) instead of only a whole folder or a single glob pattern — the field then shows “N files selected” (hover for the full file list) and the stride still applies to the chosen list.

**Transforms: Flip Y / Flip Z / Transpose** (below the Image field) apply MIDAS's `ImTransOpt` image transform to every frame this tab loads — the single preview image and every frame of the Section 2 statistical stack source alike — before any masking runs, in the fixed Flip Y  $\rightarrow$  Flip Z  $\rightarrow$  Transpose order. Receiving a calibration from Tab 2 (Section 4) pre-checks these to match the Calibrate tab's own Transforms state (still overridable here). Same codes/semantics as the Data Viewer and Calibrate tabs.

### Section 1 · Threshold mask (always applied)

`pixel  $\leq$  lower | pixel  $>$  upper`. The upper bound auto-fills from the data type on load (e.g. 1,048,575 for uint20 Eiger, 4,294,967,295 for uint32).

### Section 2 · Statistical auto-mask

Two **independently selectable** methods — enable either or both: - **Spatial outlier** — robust local-anomaly detection (5 $\times$ 5 median residual  $\rightarrow$  15 $\times$ 15 local MAD  $\rightarrow$  Z-score  $\rightarrow$  hot/dead/saturated gates). Uses a **temporal median** over the stack folder when provided (cleaner reference), else the single frame. Controls: `K $_{\sigma}$`  (6.0), `Hot` (1.5), `Dead` (0.5). - **Temporal constancy** — flags frozen pixels whose frame-to-frame std is below `Frozen  $\times$  Q75(std)` (0.05); needs a stack of  $\geq 2$  frames.

A shared **stack source + stride** feeds both (the temporal median for spatial, the frame stack for temporal constancy). The stack can be a **folder / \*.tif glob**, an explicit **multi-select file list** (via the browse menu's Files (multi-select)... entry), or a **single HDF5 file whose 3-D dataset is a time sequence of images** — for an `.h5` a **Dataset** selector appears (auto-populated, 3-D datasets preferred). *All  $\sigma$  /  $K_{\sigma}$  /  $n_{\sigma}$  fields in this tab accept any value — there is no upper limit.*

### Section 3 · Spatial spike rejection

Laplacian high-pass single-pixel-spike detector; `n $_{\sigma}$`  threshold (5.0).

### Section 3b · Cosmic-ray rejection (temporal)

Per-pixel temporal  $\sigma$ -clip across a  $\geq 3$ -frame stack; anomalies OR'd across frames. `n $_{\sigma}$`  (5.0).

### Section 4 · Calibration-based masks (need a Tab 2 result)

- **Azimuthal  $\sigma$ -clip** — flags (R,  $\eta$ ) cells deviating from the azimuthal mean.
- **Learnable mask** — differentiable per-pixel mask trained against an  $\eta$ -uniformity loss (`steps`, `lr`, `sparsity`).

### Section 5 · Post-processing

**Dilation (px)** (default 0) grows every bad pixel from Sections 1-4 (and a mask loaded via **Save / Load**) using 8-neighbor morphological dilation: at N=1 the full 3 $\times$ 3 block around each bad pixel becomes bad, at N=2 the full 5 $\times$ 5 block, and so on (a (2N+1) $\times$ (2N+1) square centered on each pixel). Applied once, on **Compute Mask** / mask load, before hand-drawn shapes are combined in — hand-drawn regions are never grown by this control.

### Draw mask tools

Interactive rectangle / oval / circle / polygon / annulus / point tools, live-draggable; **Apply shapes** → **mask** rasterises them (OR'd with the computed mask). **Clear shapes** removes them.

## Save / Load

Save the combined mask as TIFF (0 = good, 1 = bad); loading a TIFF applies immediately.

---

## 5. Tab 2 — Calibrate

**Purpose:** determine detector geometry (Lsd, BC, tilts, distortion, wavelength) from a calibrant pattern.

### Mode ribbon (leftmost strip)

Like the Data Viewer tab, a narrow vertical strip switches between **Single detector** (the view described below — unchanged) and **Hydra** (per-panel calibration for the 1-ID-E 4-panel GE detector). The two modes are independent: switching does not share data or geometry between them. As on the Data Viewer tab, **Hydra only appears when the active profile is 1-ID-E** (see §3 “Mode ribbon”, §15 “Profiles”).

### Hydra mode (4-panel GE detector)

**Purpose:** fit each of the 4 GE panels’ own detector geometry (BC, Lsd, tilts, distortion) from one calibrant dataset, using one shared “recipe” (pipeline,  $\lambda$ /pixel/calibrant, refine-parameter choice) applied to all 4 — since each GE panel is a physically separate detector, its beam centre, Lsd, and tilt are fit and shown independently.

- **Hydra data / ← Data Viewer (left panel):** the same sibling-discovery data loader as the Data Viewer tab’s Hydra mode (point at any one panel file, the other 3 are found automatically; shared frame slider; Dark / Bright / Background per panel). ← **Data Viewer** pulls the currently loaded Hydra panel path *and* every present panel’s fitted/seeded geometry (BC, Lsd, tilts, Transforms) straight from the Data Viewer tab’s Hydra page, so you don’t have to re-browse or re-pick beam centres you’ve already set up there. This mode has no stack-projection feature (like the single-detector Calibrate tab) — use **Average frames** below instead.
- **Detector & Calibrant, Threshold, Average frames, Refine parameters, Advanced (middle panel, shared across ge1–ge4):** one copy of each, identical in meaning to the single-detector tab’s own cards, applied to every panel’s fit. There is no **Multi-panel detector** (tiled sub-panel rigid-shift) group in Hydra mode — that feature refines shifts between tiles *inside* one monolithic detector’s readout, which doesn’t apply to 4 separate physical GE detectors.
- **Transforms / Initial seed (middle panel, switches with the active panel):** independent per GE panel — physical mounting orientation (Flip Y/Flip Z/Transpose) and beam centre/Lsd/tilts genuinely differ panel to panel. **Load calibration file...** seeds one panel’s BC/Lsd/tilts/ Transforms from a file (and mirrors  $\lambda$ /pixel into the shared Detector card if the file carries them); Pick BC/Pick Ring on the image seed that panel’s BC the same way as the single-detector tab. **Use manual seed** and **Feed result back to seed** are one shared choice across all 4 panels — checking/unchecking either on any one panel’s card mirrors the same state onto the other three — since it’s a single decision about *how* the LM fit should start/carry forward, not a per-panel geometry value; the seed BC/Lsd/tilt *values* themselves remain fully independent per panel.
- **Run (Run mode / Run Calibration / Abort):** **Sequential** fits one panel at a time (full per-line progress in the Log tab, same as the single-detector tab); **Parallel** starts every currently-found panel’s fit at once — faster, but each panel’s fine-grained progress prints go to the console rather than the Log tab (only start/finish/error lines appear there), since a fit’s internal progress-capture can’t safely be shared across concurrent threads.
- **Image toolbar: GE1 / GE2 / GE3 / GE4** (no Composite — calibration is inherently per-panel; see the Data Viewer tab for the windmill composite once all 4 are fitted) plus **Show rings / Corrected**, identical in meaning to the single-detector tab’s own predicted-ring overlay toggles, applied to whichever panel is active.
- **Radial Profile (bottom right):** one shared multi-curve plot — all 4 panels’ post-fit azimuthal profiles at once, same style/controls as the Data Viewer tab’s Hydra Radial Profile plot (R-bin/ $\eta$ -bin/weighting + **Re-integrate**, X-axis unit selector). **Eta vs R Cake**, **Ring Residuals** and **Results** switch to show only the currently active panel’s own 2-D cake / chart / parameter grid (Results has its own → **Send to Data Viewer**, **Save .json**, and **Save paramstest.txt**, scoped to that one panel). **Log** is shared across all 4 panels, each line prefixed [ge1] ... [ge4] .

### Data Loader panel (left)

The calibrant frame (Data + Frame index), Dark, Bright, Background and Mask are selected in the shared **Data Loader panel** (see §1). Each Dark/Bright/Background field is averaged over a chosen index range; Bright offers **Flat-field divide** or **Subtract**; all mask sources (plus the auto-added Tab 1 mask) are unioned. Dark is passed to the calibration pipeline;



bright/background are applied to the calibrant before calibration and post-calibration integration. The image preview updates live to show the dark/bright/background-corrected frame as soon as a field is picked or changed (the raw frame still feeds the actual calibration run, which applies these corrections itself). On the **1-ID-E / 20-ID-D / 20-ID-E** profiles, loading the Data file also auto-detects **pixel size** (from a `.ge1` – `.ge5` / `.vr*` / `.pxrd` filename tag) and, for an HDF5 file, **wavelength** (from its recorded beam energy) into the Detector card below — whatever isn't detected is left unchanged. (Hydra mode: the same detection runs off the loaded anchor panel file and applies to the shared  $\lambda$ /pixel fields.)

## Detector, seed & Load calibration file

The **Detector & Calibrant** card sets  $\lambda$ , pixel size(s) and detector transforms (**Flip Y / Flip Z / Transpose** — MIDAS's `ImTransOpt`, in that fixed order). Toggling a Transforms checkbox live-updates the image preview immediately, the same way Data Viewer and Mask Builder do; Pick BC / Pick Ring clicks are read straight off that (transformed) preview, so the seed beam centre always lands in the same coordinate space the fit will actually run in. The calibration run itself applies the identical transform (to the calibrant image and to Dark) right before handing the array to the pipeline, so a Flip/Transpose selection is now honoured by every pipeline mode, not just the display; the **Initial seed** card sets the LM starting point (BC\_y, BC\_z, Lsd, and tilts tx/ty/tz). **Load calibration file...** (top of the Detector card) reads a MIDAS **paramstest** `.txt`, a calibration `.json`, or a pyFAI `.poni` (auto-detected) and fills  $\lambda$ , pixel size, the seed BC + Lsd, and the Transforms checkboxes from any `ImTransOpt` lines — a fast way to start from a previous calibration or a known geometry. (The synthetic test data ships `test_data/calibration_synthetic.{json,txt,poni}` as a ready example.) ← **Data Viewer** (next to *Load calibration file...*) pulls  $\lambda$ , pixel size, Lsd, beam centre, the Data Viewer's ty/tz tilt fields, and its Transforms state straight from the Data Viewer tab into the same fields (BC and Lsd land in the seed card; ty/tz land in the seed tilt fields). A **Feed result back to seed** checkbox (on by default) copies the optimized BC / Lsd / tilts / distortion of each run back into the seed fields, so a follow-up run starts from the previous solution. *Seed tilts are honoured by the Four-stage / advanced pipelines; the One-shot / First-time paths seed tilts only if the installed backend exposes initial-tilt options.*

## Average frames

For a multi-frame source (HDF5 / folder), **Average frames into a single image** builds the mean of a frame range and calibrates on that. **start / end (0 = all)** select the range. The card is disabled for single-frame sources; the preview updates live as the options change.

## Pipeline selector

One-shot (default) · First-time · Four-stage · Bayesian (Laplace  $\sigma$ ) · Joint-cake. *For trustworthy tilt/strain, prefer Four-stage or First-time — One-shot/Bayesian can report a spurious self-compensated tilt on weakly-tilted data.*

## Refine flags

Which parameters vary: Lsd, BC, ty, tz, tx, Wavelength, plus a “Residual map” build toggle. The **Distortion (n/15)** checkbox has a companion ... button that opens a per-coefficient dialog: the 15 distortion coefficients are grouped by  $\eta$ -fold (isotropic radial + folds 1–6), and named **preset modes** (None · Isotropic only · Iso + up to 2-fold · Iso + up to 4-fold · All (15)) auto-select whole ladders. The checkbox label shows how many coefficients are selected. Picking a strict subset genuinely restricts which coefficients the LM fit refines — including on the One-shot pipeline, which transparently switches to a lower-level routine for a partial selection (its own backend otherwise only supports all-15-or-none); picking “All (15)” or leaving Distortion unchecked runs the normal One-shot path unchanged. Advanced (E-M / LM iters, device, output dir) and Multi-panel detector groups are collapsible.

## Live threshold slider

Zeroes calibration-image pixels below the slider value (background suppression for ring-finding); the preview updates instantly.

## Pick BC / Pick Ring

Click the image to seed the beam centre (single click) or fit a ring ( $\geq 3$  clicks); the Pick Ring points/fit are drawn in blue, distinct from the amber Simulate-rings overlay.

## Predicted-ring overlay (image toolbar)

After a run, the calibrant's predicted ring radii are drawn in **lime** with a red/yellow beam-centre marker. **Show rings** toggles the overlay; **Corrected** redraws the same lime rings reshaped through the fitted tilt (tx/ty/tz) instead of as plain circles, so you can see how much the geometry actually deviates from an untilted detector — the rings still look like the same thin curves, just bent, rather than a separate scattered point-cloud.



## Run / Abort

**Run Calibration** launches the worker; **Abort** terminates it and frees the slot so you can immediately start a new run (the calibration is one uninterruptible library call, so abort hard-terminates the worker thread rather than waiting).

## Results (right panel, bottom tabs)

Radial Profile (with ring markers and an **Azim. avg** selector — see Tab 4), **Eta vs R Cake**, Ring Residuals bar chart, **Results**, and Log. Integration runs automatically after calibration. The **Eta vs R Cake** tab shows the same integration as a 2-D heatmap instead of collapsed to a 1-D curve — R (px) on the X-axis,  $\eta$  (°) on the Y-axis, intensity as color — the routine “cake” visualization for area-detector diffraction data; it has its own Log/colormap/vmin%/vmax% controls (same conventions as the main image viewer) and updates on every **Re-integrate**. The **Results** tab shows the parameter set exactly as it is written to `paramtest.txt` (Lsd, BC, tx/ty/tz, the distortion coefficients, Parallax, Wavelength, px, NrPixelsY/Z, RhoD, SpaceGroup, LatticeConstant, and any `ImTransOpt` codes from the Transforms checkboxes) as **plain text laid out in multiple columns** so the wide-but-short panel stays readable — no table widget. The distortion `p0–p14` slots are labelled with their coefficient names (iso\_R2, a1, phi1, ...), and only the coefficients actually selected for that run are listed (not all 15), so the display matches what you asked to refine; the saved `paramtest.txt / .json` exports still carry every slot's real value, including any held-fixed value carried over from a prior calibration. A strain/timing line sits below. → **Send to Data Viewer** pushes the *full* calibrated geometry ( $\lambda$ , pixel size, Lsd, beam centre, **tilts, distortion, and the Transforms state**) into the Data Viewer tab — where it drives the tilt/distortion-aware radial integration, not just circle binning, and sets Data Viewer's own Flip/Transpose checkboxes to match — the reverse of the Data Viewer's “→ Send geometry to Calibrate”. The bottom tab area is fully resizable (drag the horizontal splitter) and the Log fills its tab.

## Export

**Save calibration.json** and **Save paramtest.txt** (standalone or from a template). Both carry the Transforms checkboxes' `ImTransOpt` codes (one `ImTransOpt <code>` line per checked transform in the `.txt`; an `im_trans` list in the `.json`), so reloading either file elsewhere restores the same detector orientation.

---

## 6. Tab 3 — Calibration Refinement

**Purpose:** optional post-calibration geometry refinement against an  **$\eta$ -uniformity** criterion (rings should be azimuthally uniform). Uses a **derivative-free Nelder-Mead** optimiser (the differentiable integrator returns NaN geometry gradients at the beam-centre singularity on this build).

The sample frame and Dark/Bright/Background/Mask are selected in the shared **Data Loader panel** (see §1) — the refinement runs on the corrected image. Middle-panel controls: calibration source (Tab 2 result, or start-from/continue radios), parameters to refine (BC\_y, BC\_z, Lsd, ty, tz), optimiser + iterations. **Run Refinement** shows a live loss curve; **Apply** broadcasts the refined geometry downstream. If Apply is never clicked, the Tab 2 result flows through unchanged.

---

## 7. Tab 4 — Batch Integrate

**Purpose:** integrate a stack of frames into 1-D profiles using the calibrated geometry and optional corrections.

### Mode ribbon (leftmost strip)

Like the Data Viewer and Calibrate tabs, a narrow vertical strip switches between **Single detector** (the view described below — unchanged) and **Hydra** (per-panel integration for the 1-ID-E 4-panel GE detector). The two modes are independent: switching does not share data or geometry between them. As on the other two tabs, **Hydra only appears when the active profile is 1-ID-E** (see §3 “Mode ribbon”, §15 “Profiles”).

### Hydra mode (4-panel GE detector)

**Purpose:** integrate each of the 4 GE panels' own frame stack with its own independently fitted geometry, using one shared integration “recipe” (kernel, bins, corrections, monitor normalisation, output format) applied to all 4 — since each GE panel is a physically separate detector with its own geometry, but the same beam and the same choice of what to compute.

- **Hydra data (left panel):** the same sibling-discovery data loader as the other Hydra pages (point at any one panel file, the other 3 are found automatically), but in **streaming** form for batch runs — a shared frame **range + stride** (frames are synchronized across panels, so one range applies to all) instead of a frame navigator, Dark/Bright/ Background per panel,

and an independent **Mask** per panel (its own file/folder sources, unioned — no cross-panel auto-discovery, since mask files are physically panel-specific and may not follow the `ge{n}` naming convention data files do).

- **Integration, Physics corrections, Monitor normalisation, Output (middle panel, shared across ge1–ge4)**: one copy of each, identical in meaning to the single-detector tab's own cards, applied to every panel's run. Each panel writes its output into its own `ge{n}/` subfolder under the shared output directory, so the 4 panels' files never collide. There is no Drift correction or live **MONITOR** folder-watch in Hydra mode (both single-detector-only for now).
- **ge1 – ge4 toggle + Calibration source / values / progress (middle panel, switches with the active panel)**: each panel picks **From Calibrate tab** (auto-populated the moment that panel's fit finishes on the Calibrate tab's own Hydra page — no action needed here) or **From file** (a calibration `.json`, MIDAS `paramstest.txt`, or pyFAI `.poni`, same auto-detection as the single-detector tab); the **Calibration values** grid and a compact progress bar are per panel.
- **Run (Run mode / Start Integration / Abort)**: **Sequential** integrates one panel at a time; **Parallel** starts every currently-found panel's run at once. Abort stops every running panel after its current frame, keeping frames already written.
- **Waterfall / Stacked profiles (right panel, switches with the active panel)**: each panel has its own pair, same controls as the single-detector tab's own viewers. A shared **Log** below is prefixed `[ge{n}]` per line so all 4 panels' activity can be read from one place.

## Data Loader panel (left)

The streaming **Data** source (folder/glob or HDF5 dataset) with **frame range + stride**, plus Dark/Bright/Background and Mask, are selected in the shared **Data Loader panel** (see §1). Dark/bright/background are applied per frame; all mask sources are unioned.

## Calibration source (middle)

- **From Tab 2** — the calibration (or refined) result.
- **From file** — a **calibration** `.json`, **MIDAS** `paramstest.txt`, or **pyFAI** `.poni` (auto-detected; both GUI and MIDAS-pipeline json key styles supported). Entering a path auto-selects this option.

## Calibration values (middle, read-only)

A **Calibration values** card shows the geometry actually in use —  $\lambda$ , Lsd (mm), BC\_y/BC\_z (px), tilts tx/ty/tz (°), pixel sizes, detector size, and a distortion summary (number of non-zero coefficients). It repopulates whenever a calibration arrives from Tab 2 (or Tab 3 refinement, or the Data Viewer via Tab 2) **and** when a calibration file path is entered, parsing the file directly so you can confirm the numbers before running. A note line reports the active source (or any file-read error).

## Integration

| Field                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kernel                        | Hard (fastest) · Subpixel K=2 · Subpixel K=4 · Polygon (exact).                                                                                                                                                                                                                                                                                                                                                                                       |
| R bin (px) / $\eta$ bin (°)   | Radial and azimuthal bin sizes.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Azim. avg</b>              | How the ( $\eta$ , R) cake becomes a 1-D profile: <b>Pixel-weighted</b> (default) $\Sigma(\text{mean} \cdot \text{count}) / \Sigma(\text{count})$ — independent of $\eta$ -bin size and robust to partial azimuthal coverage / <b>off-detector beam centres</b> ; or <b><math>\eta</math>-bin mean (legacy)</b> — the unweighted mean of per- $\eta$ -bin means, which can distort with a coarse $\eta$ bin when the beam centre is off the detector. |
| Per-bin variance ( $\sigma$ ) | Error model poisson / azimuthal / hybrid (ignored when corrections are on $\rightarrow \sigma = \sqrt{l}$ ).                                                                                                                                                                                                                                                                                                                                          |
| Q-uniform bins                | Integrate in R then rebin onto a uniform-Q grid (Qmin, Qmax, $\Delta Q$ ).                                                                                                                                                                                                                                                                                                                                                                            |

## Physics corrections

Polarization and solid-angle (pixel-domain, via `integrate_with_corrections`).

## Monitor normalisation

Divide each processed frame's profile/ $\sigma$  by a per-frame scalar from a text file (one value per *processed* frame).

## Drift correction (long scans)

Per-frame geometry interpolated from an anchor JSON ( `{frame_idx: {Lsd, BC_y, BC_z}}` ) with spline/linear/constant parametrization; **Fit trajectory** builds it. Each frame is then integrated with its own geometry.

## Run / Abort / Clear

**Start Integration / Abort.** Abort first asks the worker to stop cleanly between frames (keeping frames already written); if it does not stop promptly it force-terminates and frees the slot. Completed frames' output files are preserved. **Clear results** removes the profiles/plots computed **this session** for the current data (waterfall, stacked profiles, and the integrated-frame tracking) so a fresh integration can start — it stops any active monitor but does **not** delete raw data or any files already written to disk.

## Live folder monitoring (MONITOR)

The **MONITOR** button at the bottom of the left Data Loader panel (folder/glob sources only) starts a live watch: while active it turns **green** and the GUI polls the data folder for **new** TIFF frames, integrating each one **as it appears** and adding it to the display. It does **not** re-run the whole batch — it reuses the already-built **detector map** (the binning geometry / pixel-count cakes; reused from a prior *Start Integration* run when the calibration, kernel, bins, mask and folder are unchanged, or built once on first use) and integrates **only the new files** (tracked by frame id, so frames already shown are skipped). New frames honour the current kernel, corrections, Dark/Bright/Background, mask and Q-uniform settings, and are saved to the output folder when a 1-D format is selected. Click MONITOR again to stop; starting a fresh *Start Integration* also stops it. (Distinct from **Monitor normalisation** above, which divides profiles by a scalar file.)

## Output formats

CSV (R,l, $\sigma$ ) · XYE (2 $\theta$ ) · FXYE (centideg) · DAT (Q) · HDF5 (full stack) · 2D-CSV ( $\eta \times R$  cake). Right panel: live **Waterfall** and **Stacked profiles** — both have an **x** selector to show the axis in **R (px)** / **2 $\theta$  (°)** / **Q (Å<sup>-1</sup>)** (converted from the run's calibration).

The **Stacked profiles** view is a publication-quality plot: each curve tags its **source file name inline, just below the curve at its left edge** (toggle with the *Labels* checkbox; an optional corner *Legend* is also available). A **theme** selector switches between two saved presets: - **White (publication)** (*default*) — white background, **point + line** markers, a colour-blind-friendly categorical palette (matplotlib *tab10*), a boxed frame and a light grid — ready to drop into a paper. - **Dark** — the classic on-screen look (dark background, line-only, vivid golden-angle colours).

An **x** selector plots the axis in **R (px)**, **2 $\theta$  (°)** or **Q (Å<sup>-1</sup>)** (converted from the run's calibration). The **Grid** checkbox (off by default) toggles the horizontal + vertical grid. The *spacing* box shifts successive curves vertically (0 = overlay). Top-right controls adjust the **line width** ( - line + ), **symbol size** ( - sym + , for the point+line markers — also turns markers on if a line-only theme is active) and **label font size** ( - font + ).

---

## 8. Tab 5 — Corrections & Physics (*work in progress*)

Preview physics corrections on a single frame (auto-loads on browse). Pixel-domain: polarization, solid angle, empty subtraction. Profile-domain: cylindrical absorption ( $\mu R$ ), Compton subtraction (composition:fraction). **Compute corrected profile** overlays corrected vs uncorrected and shows the correction factor vs 2 $\theta$ .

Also hosts **per-pixel gain training (LearnableGain)**: from a clean reference frame and a drifted frame, learn a spatial gain map  $g_i = 1 + \text{scale} \cdot r_i$  by minimising  $\text{MSE}(\text{profile}) + \text{unity} \cdot \sum (g-1)^2 + \text{smooth} \cdot \text{TV}(g)$  ; save as NPZ and apply with `corrected = raw / gain_map`.

---

## 9. Tab 6 — PDF Analysis

Polyatomic **total-scattering** workflow, powered by the `midas_pdf` backend:  $I(Q) \rightarrow$  Faber-Ziman structure function  $S(Q) \rightarrow$  pair-distribution  $G(r)$  (Stage 1: real composition-weighted normalization, Compton subtraction, end-to-end  $\sigma$  propagation, optional differentiable scale/background refinement) **plus** a full Stage 2-3 corrections/analysis surface — empty-cell/Paolman-Pings absorption subtraction, detector-efficiency correction, absolute (electron-unit) normalization, differentiable multiple-scattering correction, a fluorescence diagnostic, CIF-driven small-box structure refinement (PDFfit-style, error-aware), and  $\Delta$ -PDF significance testing between two saved reductions. The tab is organized as a **4-tab left panel** (Data & Reduction / Corrections / Structure Fit /  $\Delta$ -PDF) driving a **4-tab right panel** (Reduction / Structure Fit /  $\Delta$ -PDF / Log).

Ships with a dedicated, ready-to-run dataset at `test_data/test_pdf/` (real beamline frames + pre-integrated  $I(Q)$  for Ni/CeO<sub>2</sub>/IPA/Kapton/air-scatter, a rasterized beamstop mask, a MIDAS calibration file, and an authored `Ni.cif`) — kept out of git via `.gitignore` (~320 MB raw frames) but present on this machine, so the tab opens ready to run against real data with no setup.

## Background — what $I(Q)$ , $S(Q)$ and $G(r)$ mean

PDF analysis is three transforms that move from *what the detector measured* to *where the atoms are*. They are exactly the three stacked plots (top → bottom).

- **$I(Q)$  — measured scattered intensity.** Intensity vs. momentum transfer  $Q = 4\pi \cdot \sin(\theta)/\lambda$  (Å<sup>-1</sup>), a wavelength-independent rescaling of the angle  $2\theta$ ; high  $Q$  = wide angle = fine spatial detail. Obtained by azimuthally integrating the detector rings to a 1-D curve.  $I(Q)$  is **not** pure structure — it mixes the interference (coherent) signal with big smooth backgrounds: per-atom self-scattering (form factors  $\langle f^2 \rangle$ ) and inelastic **Compton** scattering.
- **$S(Q)$  — structure function.**  $I(Q)$  with the self-scattering and Compton removed and normalized away (the **Faber-Ziman** step), leaving only interference:  $S(Q) = [I_{\text{coh}} - (\langle f^2 \rangle - \langle f \rangle^2)] / \langle f \rangle^2$ , where  $\langle f \rangle$ ,  $\langle f^2 \rangle$  are composition-weighted atomic form factors (hence the **Composition** input).  $S(Q)$  oscillates about **1** and → 1 at high  $Q$  where correlations wash out (the dashed guide line).  $F(Q) = Q \cdot [S(Q)-1]$  is the same thing re-weighted by  $Q$  to emphasize the structurally rich high- $Q$  oscillations.
- **$G(r)$  — reduced pair distribution function.** The real-space answer: a histogram of interatomic distances, via a sine Fourier transform  $G(r) = (2/\pi) \cdot \int Q \cdot [S(Q)-1] \cdot \sin(Qr) \, dQ$ . A **peak at  $r$**  means many atom pairs are separated by that distance; the **first peak is the nearest-neighbour bond** (Ni ≈ 2.49 Å, CeO<sub>2</sub> Ce–O ≈ 2.34 Å), later peaks are further coordination shells. Below the first bond  $G(r) = -4\pi\rho_0 r$  (nothing can be closer) — a constraint that needs the number density  $\rho_0$ . Because it uses *total* scattering (Bragg **and** diffuse), the PDF captures local structure even in disordered / nanoscale / amorphous materials, unlike a Bragg-only Rietveld fit.

```

detector image / I(Q) file
| azimuthal integration
▼
I(Q)  raw intensity = interference + form factors + Compton      (top plot)
| subtract Compton, subtract Laue ( $\langle f^2 \rangle - \langle f \rangle^2$ ), divide by  $\langle f \rangle^2$  (needs composition)
▼
S(Q)  pure interference, oscillates about 1                      (middle plot)
| Fourier sine transform over [Qmin, Qmax] with a window (Lorch)
▼
G(r)  interatomic distances; peaks = coordination shells          (bottom plot)

```

The **±1σ band** on  $G(r)$  is the counting uncertainty  $\sigma$  propagated from  $I(Q)$  through every step, so real peaks can be told from noise. The **G/g/T/R** family (Output selector) is the same information re-weighted: **G** oscillates about 0 (refinement standard); **g** → 1 at large  $r$ ; **T** is the total correlation; **R** is the radial distribution whose *peak area = coordination number*.  $g/T/R$  all need  $\rho_0$ .

## Left tab 1 — Data & Reduction

**$I(Q)$  source** (choose one): - **Integrate detector frame** — a calibrated frame is integrated (hard/polygon binning, Poisson  $\sigma$ ) and mapped to  $Q$ . Uses the Tab 2 calibration ( $\lambda$ , geometry) and an optional local **Mask** (`.tif`, nonzero = masked — same convention as Tab 1, loaded independently so this tab needs no cross-tab wiring to run). - **Load  $I(Q)$  file** — a pre-integrated 2- or 3-column  $Q$ ,  $I$ ,  $\sigma$  text/CSV (comment/header lines tolerated). The tab **opens on this source by default**, pointed at `test_data/test_pdf/iq/04_iq_Nickel.csv` — just press **Compute  $G(r)$**  for the Ni PDF (first shell ≈ 2.5 Å). The same folder also has CeO<sub>2</sub>, IPA, Kapton (container), and air-scatter  $I(Q)$  at  $\lambda$  0.1839 Å for use as backgrounds or empty-cell references (see Corrections, below).

**Calibration.** The geometry/wavelength comes from Tab 2 automatically, or use **Load calibration file...** to read a MIDAS `paramtest.txt`, a `.json`, or a pyFAI `.poni` (auto-detected). This is required for *Integrate detector frame* and also sets  $\lambda$  used by *Load  $I(Q)$  file* mode.

**Sample.** Composition (e.g. `Ni` or `C:3,H:8,O:1`; ions like `Ni2+` allowed), number density  $\rho_0$  (atoms/Å<sup>3</sup>; needed for refinement and for  $g/T/R$ ), wavelength  $\lambda$  (auto-filled from calibration), and a **Compton subtraction** toggle.

**Background subtraction (empty-cell).** Optional. Loads a second  $I(Q)$  file (e.g. the Kapton or air-scatter references) and subtracts it from the sample: **Manual scale** (a fixed transmission factor  $s$ ,  $I_{\text{corr}} = I - s \cdot I_{\text{empty}}$ ) or **Fit high-Q** (least-squares  $s$  over a chosen high- $Q$  window). Enabling **Paalman-Pings cylinder-in-cylinder correction** replaces the flat scale

with the Q-dependent self-/container-absorption ratio, using sample/container linear attenuation  $\mu$  ( $1/\mu\text{m}$ ) and cylinder radii; **Estimate  $\mu$  from composition** fills  $\mu$  from the sample/container composition,  $\lambda$ , and container density automatically.

**Normalization.** Optional **Refine scale + background** (L-BFGS) — reveals a background polynomial degree (0–3), a low-r cutoff  $r_{\text{min}}$  (Å), and an iteration count. Refinement requires  $\rho_0 > 0$ . It fits scale and a smooth  $b(Q)$  against two model-free constraints: high-Q  $\langle S \rangle \rightarrow 1$  and  $G(r) = -4\pi\rho_0 r$  below  $r_{\text{min}}$ .

**Q range / r range + FT / Output.** Unchanged from Stage 1: Qmin/Qmax trim, rmin/rmax/ $\Delta r$  + window (Lorch/none) + binning (hard/polygon), and the bottom-plot convention family — **G(r)** (reduced PDF), **g(r)** (pair distribution), **T(r)** (total correlation), or **R(r)** (radial distribution, peak integral = coordination number); g/T/R need  $\rho_0$ . Middle plot toggles between **S(Q)** (with the S=1 guide) and **F(Q)=Q(S-1)**.

**Save G(r)** writes a three-column  $r, G(r), \sigma$  file (diffpy-CMI / PDFgui / RMCProfile compatible); **Save S(Q)** writes  $Q, S(Q)$ .

## Left tab 2 — Corrections

All four stages are opt-in checkboxes; leaving all off reproduces Stage-1 behavior exactly.

- **Detector efficiency** — divides  $I(Q)$  by the sensor's quantum efficiency  $\eta(Q)$  for a given material/thickness (e.g. 500  $\mu\text{m}$  Si), computed from the photoelectric absorption at each Q's implied path length. At hard X-ray energies a thin high-Z-poor sensor can be only a few percent efficient, so this legitimately amplifies  $I(Q)$  severalfold — that is physically expected, not a bug.
- **Multiple scattering** — differentiable cylinder-transport correction (`slab_transport_ms / ms_background_on_grid`): given an effective optical depth  $\tau$  (from  $\mu \times R$ ,  $\mu$  either auto-estimated from composition/density or set manually), a single-scattering albedo, and a Q grid, computes and subtracts an MS background before normalization. The Log reports the median MS fraction  $\beta$ .
- **Absolute (electron-unit) normalization** — anchors the mean  $I(Q)$  over a high-Q window to the composition's  $\langle f^2 \rangle + \langle S_{\text{inc}} \rangle$  baseline, putting  $I(Q)$  on an absolute per-electron scale (needed before meaningfully comparing intensities across samples/geometries).
- **S(Q) tail-flatten** — a **display-only** PDFgetX3-style iterative MAD-clipped polynomial baseline flatten over a high-Q window; toggled on the Reduction plot via **Show tail-flattened S(Q)**. It never feeds back into  $G(r)$  — the reduction always uses the un-flattened  $S(Q)$ .
- **Fluorescence diagnostic** — **Check fluorescence** reports which sample/container element K/L emission lines fall near the incident energy (a source of spurious background near an absorption edge), via `expected_fluorescence / fluorescence_report_sample_and_container`.

## Left tab 3 — Structure Fit

CIF-driven (or manually-specified lattice + atom table) small-box structure refinement against the observed  $G(r)$ , in the style of PDFfit/PDFgui:

- **Crystal** — load a `.cif` (e.g. `test_data/test_pdf/structures/Ni.cif`, authored FCC Ni, space group Fm-3m #225,  $a=3.524$  Å) or build one manually (lattice  $a/b/c/\alpha/\beta/\gamma$ , space-group number, an atom table with add/remove rows).
- **Fit range + parameters** — pair-list cutoff (`build_pair_list r_max`), the `[fit rmin, fit rmax]` window actually fit, a  $\sigma$  **inflate** factor (real beamline  $G(r)$  is often dominated by shape-mismatch systematics rather than counting noise, so  $\sigma$  can be scaled up before fitting — exposed as a tunable, not hardcoded), initial guesses for lattice constant / isotropic ADP (`u_iso`) / scale, an optional background polynomial order, and optimizer steps/lr/ posterior-sample count.
- **Run structure fit** calls `refine_structure` (gradient-based, PyTorch) and reports fitted values  $\pm$  uncertainty,  $\chi^2/\text{ndof}$ , and observed/model/residual curves on the right panel's **Structure Fit** tab.

## Left tab 4 — $\Delta$ -PDF

Significance testing between two saved  $G(r)$  snapshots (e.g. before/after a correction, or two temperatures): **Save current result as State A/B** each capture  $(r, G, \sigma)$  from the last Compute; **Compute  $\Delta G(r) = B - A$**  calls `delta_pdf / significant_mask` and plots  $\Delta G(r)$  with an  $n\text{-}\sigma$  band, marking points that exceed the chosen threshold as a red scatter overlay — a fast way to see *where* two reductions genuinely differ vs. where the difference is just noise.

## Right panel

- **Reduction** —  $I(Q)$  (+ background overlay),  $S(Q)/F(Q)$  (with the tail-flatten toggle), and the selected  $G/g/T/R$  with a shaded  $\pm 1\sigma$  band — same three-stacked-plot layout as Stage 1.
- **Structure Fit** — observed  $G(r)$  ( $\pm 1\sigma$  band) with the fitted model overlaid, a `(obs-calc)/ $\sigma$`  residual sub-plot, and a fitted-parameter table with  $\chi^2/\text{ndof}$ .



- **Δ-PDF** —  $\Delta G(r)$  with its uncertainty band and significant-point scatter, plus an “N/Ntotal points > no” summary label.
- **Log** — shared run log for reduction, corrections, structure fit, and  $\Delta$ -PDF, in the same `LogPanel` used elsewhere.

## Functions behind the tab

The tab is a thin UI over two background workers and the `midas_pdf` backend.

**UI** — `PDFTab` ( `midas_gui/tab_pdf.py` )

| Method                                                                                                                 | Role                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>_build_ui</code>                                                                                                 | Builds the 4-tab left panel and 4-tab right panel described above.                                                                                                     |
| <code>set_calibration(result, source)</code>                                                                           | Receives a Tab-2 result (or a loaded geometry) → sets $\lambda$ and enables Compute.                                                                                   |
| <code>_load_calib_file</code>                                                                                          | Loads a <code>paramtest/.json/.poni</code> → builds a calibration result → <code>set_calibration</code> .                                                              |
| <code>_load_img</code> / <code>_load_mask</code>                                                                       | Loads a detector frame / local <code>.tif</code> mask for <i>Integrate detector frame</i> mode.                                                                        |
| <code>_estimate_mu</code>                                                                                              | Fills sample/container $\mu$ ( $1/\mu\text{m}$ ) from composition + $\lambda$ (+ container density) for the Paalman-Pings correction.                                  |
| <code>_check_fluorescence</code>                                                                                       | Calls <code>fluorescence_report_sample_and_container</code> and reports the result in the Log + a message box.                                                         |
| <code>_run</code>                                                                                                      | Validates inputs, assembles the full Stage-1 + Stage-2/3 <code>cfg</code> dict, and starts a <code>PDFWorker</code> .                                                  |
| <code>_on_done</code> / <code>_redraw_mid</code> / <code>_redraw_bottom</code>                                         | Draw $I(Q)+bg$ , the $S(Q)\leftrightarrow F(Q)$ toggle (incl. tail-flattened), and the $G/g/T/R$ curve with its $\pm 1\sigma$ band.                                    |
| <code>_collect_fit_cfg</code> / <code>_run_structure_fit</code> / <code>_on_fit_done</code> / <code>_redraw_fit</code> | Build the CIF/manual crystal + fit-parameter <code>cfg</code> , run <code>PDFStructureFitWorker</code> , and draw the observed/model/residual plots + parameter table. |
| <code>_save_state</code> / <code>_run_delta_pdf</code> / <code>_redraw_delta_pdf</code>                                | Snapshot $(r, G, \sigma)$ into State A/B and compute/plot <code>delta_pdf/significant_mask</code> .                                                                    |
| <code>_save_gr_file</code> / <code>_save_sq_file</code>                                                                | Export $r, G, \sigma$ and $Q, S$ .                                                                                                                                     |

**Workers** — `midas_gui/workers.py` run off the GUI thread:

| Worker / function                                                                                      | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>PDFWorker._acquire_iq</code>                                                                     | Produces $(q, I, \sigma)$ — either by integrating the frame (image mode) or by reading a file ( <code>_load_iq_file</code> , to comma/space and 2- or 3-columns).                                                                                                                                                                                                                                                                                                                        |
| <code>PDFWorker.run</code>                                                                             | Trims to $[Q_{\min}, Q_{\max}]$ , builds <code>Composition</code> , then runs the opt-in Stage 2-3 stages in order — background sub (manual or fit-scale, optional Paalman-Pings), detector efficiency, absolute normalization, multiple scattering the existing refine/non-refine normalization, then an optional display-only $S(Q)$ tail-flatten; emits <code>q, Iq, background, S, S_flat, Fq, r, Gr, sigma_Gr, Gr_family, scale, bg_coef, refine_loss, bg_scale_used, ms_bet</code> |
| <code>_fit_subtraction_scale</code> / <code>_absolute_normalize</code> / <code>_flatten_sq_tail</code> | Hand-written recipe helpers (numpy-only, not part of <code>midas_pdf</code> itself) for the fit-scale empty-cell mode, abso normalization, and the tail-flatten display transform.                                                                                                                                                                                                                                                                                                       |
| <code>PDFStructureFitWorker.run</code>                                                                 | Builds a <code>Crystal</code> (from CIF via <code>read_cif_to_crystal</code> , or manually from <code>Lattice/SpaceGroup.from_number</code> / builds the pair list ( <code>build_pair_list</code> ), masks $(r, G, \sigma)$ to the fit window (with <code>sigma_inflate</code> ), calls <code>refine_structure</code> , and emits <code>fitted, uncertainty, chi2_reduced, history, r_fit, G_obs, G_calc, sigma_fit, posterior, cov</code> .                                             |

Image-mode integration reuses the shared core `_build_spec`, `build_geom`, `integrate_frame`, `axis_conversions` (same path as the other tabs); geometry-file parsing uses `geometry_fields_from_file` / `result_ns_from_geometry_file` (`midas_gui/helpers.py`).

**Reduction** — `midas_pdf` via `midas_gui/pdf_backend.py` (re-exported symbols):

| Function                                                                                                             | Does                                                                                                                                                                  | Maps to plot             |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <code>Composition(fractions, number_density)</code>                                                                  | Composition layer:<br>$\langle f \rangle$ , $\langle f^2 \rangle$ form-factor averages, Laue term $\langle f^2 \rangle - \langle f \rangle^2$ , and per-atom Compton. | —                        |
| <code>faber_ziman_S(I, q, comp, ...)</code>                                                                          | $I(Q) \rightarrow S(Q)$ with $\sigma$ (subtract Compton/Laue, divide by $\langle f \rangle^2$ ).                                                                      | $S(Q)$                   |
| <code>structure_function_F(q, S)</code>                                                                              | $F(Q) = Q \cdot (S-1)$ .                                                                                                                                              | $F(Q)$                   |
| <code>fourier_sine_transform(q, S, r, window)</code>                                                                 | $S(Q) \rightarrow G(r)$ with $\sigma$ (Lorch / none window).                                                                                                          | $G(r)$                   |
| <code>i_of_q_to_Gr(q, I, comp, r, ...)</code>                                                                        | End-to-end $I(Q) \rightarrow G(r)$ (composes the two above); the default Compute path.                                                                                | $S(Q) + G(r)$            |
| <code>refine_normalization(q, I, comp, r, ...)</code>                                                                | L-BFGS fit of scale + background polynomial against high-Q $\langle S \rangle \rightarrow 1$ and low-r $G = -4\pi p_0 r$ .                                            | refined $S(Q)/G(r) + bg$ |
| <code>pair_distribution_g /</code><br><code>total_correlation_T /</code><br><code>radial_distribution_R</code>       | Convention family from $G(r) + p_0$ .                                                                                                                                 | $g/T/R$                  |
| <code>paalman_pings_cylinder_in_cylinder /</code><br><code>paalman_pings_cell_only</code>                            | Q-dependent self-/container-absorption correction for empty-cell subtraction.                                                                                         | background               |
| <code>apply_detector_efficiency /</code><br><code>detector_efficiency /</code><br><code>linear_attenuation_um</code> | Detector-efficiency correction + the $\mu$ used by it and by Paalman-Pings/MS.                                                                                        | $I(Q)$                   |
| <code>cylinder_effective_tau /</code><br><code>slab_transport_ms /</code><br><code>ms_background_on_grid</code>      | Differentiable cylinder multiple-scattering transport $\rightarrow$ MS background.                                                                                    | $I(Q)$                   |
| <code>expected_fluorescence /</code><br><code>fluorescence_report_sample_and_container</code>                        | Predicted fluorescence emission lines near the incident energy.                                                                                                       | diagnostic only          |
| <code>read_cif_to_crystal / build_pair_list /</code><br><code>refine_structure</code>                                | CIF loading, pair-list construction, and the gradient-based small-box structure fit.                                                                                  | Structure Fit tab        |
| <code>delta_pdf / significant_mask</code>                                                                            | Difference + $\sigma$ -significance test between two $G(r)$ snapshots — <b>require <code>torch.Tensor</code> inputs</b> , not numpy arrays.                           | $\Delta$ -PDF tab        |

**Packaging note.** `midas_pdf` is the public PyPI package (pinned in `environment.yml` / `pyproject.toml`), imported directly through `midas_gui/pdf_backend.py`. Earlier versions of the GUI carried a vendored copy under `midas_gui/_vendor/` plus a `midas_hkls.absorption` compatibility shim, needed before `midas-pdf` was published and while `midas-hkls` was pinned below the release that added `absorption`; both were retired once `midas-hkls`  $\geq 0.5.0$  and `midas-pdf` were available. Deliberately **not** re-exported (out of scope, see `.context/ROADMAP.md`): the Bayesian SVI/NUTS posterior path, RMC big-box refinement, and the non-differentiable Monte-Carlo multiple-scattering variants.

---

## 10. Tab 7 — Texture / Pole Figure (*work in progress*)

Per-ring azimuthal analysis for preferred orientation. Controls: calibration source, sample frame,  $R/\eta$  bins, ring index,  $\chi$  (tilt) /  $\phi$  (rotation). **Compute Pole Figure** integrates to a cake and extracts  $I(\eta)$  at the selected ring; the right panel shows the stereographic pole figure and the raw  $I(\eta)$ . **Save pole figure (.pol)** exports POPLA format.

---

## 11. Pump Probe (time-resolved / TR-XRD)

Analyses time-resolved (pump-probe) diffraction the way the TRR group does: a folder of raw detector frames is pooled by a filename prefix, the pump-probe **delay** is parsed from each name, every frame is integrated to  $I(q)$  with the **MIDAS engine** (the same core as Batch Integrate, driven by a calibration), repeats at each delay are averaged, and a reference (mean of the pre-time-zero / negative delays) is subtracted to give  $\Delta I(q, \text{delay})$ .

**File-naming / pooling.** Frames follow `PREFIX-<fshw>fshw<delay>delay<id>.tif`, where `PREFIX` (e.g. `Ex01_Sa01_Sc17` = Experiment / Sample / Scan) is one opaque grouping key. **Scan folder** globs `PREFIX*.tif` in the loaded folder, parses `fshw` and `delay` (seconds) from each name, and reports how many frames and unique delays were found. The delay sign is flipped on load so positive delay = after the pump.

The tab uses the same three-panel layout as Batch Integrate: **data loader** (left), **settings** (middle), **plots** (right).

**Left — data loader** (the shared loader panel, as on Calibrate / Batch): - **Data** — the folder of raw frames. Defaults to the bundled TRR test set in `test_data/trr_s7id/pump_probe_BT0/detimages/` (125 Pilatus2M frames; a large, git-ignored local asset), so the tab is populated on open. An index range / stride can subset the pooled frames. - **Dark / Bright / Background** — per-frame field corrections (compute each field, then it is applied to every frame before integration). - **Mask** — a mask file and/or the mask from the Mask Builder tab. For the TRR data the tab pre-loads `invert_mask.tif` (0 = valid pixel, 1 = bad pixel / module gap), which matches MIDAS's convention that non-zero pixels are masked out.

**Middle — settings.** - **Calibration source** — *From Tab 2* (the live calibration) or *From file* (`calibration.json` / `paramstest.txt` / `.poni`). Same convention as Batch Integrate. For the shipped TRR test data the field defaults to a ready MIDAS `paramstest` (`Ex01_Sa01_Sc17_midas.txt`, converted from the dataset's pyFAI/Fit2D geometry), so the tab integrates out of the box. - **Data pooling (TRR)** — the pooling **prefix + Scan folder** (the folder itself comes from the loader on the left). - **Integration** — kernel,  $R/\eta$  bins, the plot axis ( $Q / 2\theta / R$ ), and optional Q-uniform binning. Identical engine and options to Batch Integrate. - **Physics corrections** — polarization / solid-angle. - **Pump-probe options** — the reference-delay set (defaults to all negative delays; multi-select to override), optional per-pattern normalization over a q-window, the  $\Delta I$  colour range (auto or fixed  $\pm$ ) and the diverging colormap.

**Views (right panel).** Every axis shows **real physical values**, not indices — the radial axis is in  $Q$  ( $\text{\AA}^{-1}$ ),  $2\theta$  ( $^\circ$ ) or  $R$  (px) per the plot-axis selector, and delays are in seconds. 1.  **$\Delta I$  heatmap** —  $\Delta I$  over the radial axis (a true, continuous  $Q/2\theta/R$  axis) versus delay, diverging colour centred at zero, with a labelled  $\Delta I$  colour-bar and the reference  $I(q)$  lineout beside it (shared radial axis). Delays span several decades and are irregular, so they are laid out as columns labelled with their real delay values. 2.  **$\Delta I$  vs q** — one curve per delay, coloured along a rainbow by delay. With  $\leq 8$  delays each curve is named in the legend; with more, a continuous **delay colour-bar** (min  $\rightarrow$  max, in seconds) replaces the legend. 3. **Kinetics** —  $\Delta I$  versus delay for user-defined q-bands (**Add band** over a q range); x-axis as real delay (**linear**), **log** (post- $t_0$  delays, natural for the decade-wide delay range), or **rank**. 4. **Mean patterns** — the averaged  $I(q)$  at each delay plus the dashed reference, with an optional  **$\pm 1\sigma$  band** (spread of  $I(q)$  across delays) and a **Log Y** toggle to reveal weak features across the full dynamic range; a signal-level / stability check.

All views use a publication-style white theme with large, clearly-labelled axes, readable legends and a subtle grid. A toolbar above the plots sets the **draw** mode (lines / lines+points / points) and the **line**, **point (sym)** and **font** sizes (the  $-/+$  groups), as on the Batch Integrate tab. Pan and zoom are bounded to the data so you cannot lose the plot area.

Integration runs off the GUI thread (`PumpProbeWorker`) with a progress bar and **Abort**. There is no peak fitting — peak position / width / area are read from the plots, matching the reference workflow.

---

## 12. Results & Export (*work in progress*)

Session summary + one-click export. Checkboxes select which products (calibration.json, paramstest.txt, mask.tif, integrated profiles, G(r), pole figures, session log) to copy to an output directory. A provenance block (package versions, geometry hash, mask fraction, correction flags) can be copied to the clipboard for a Methods section.

---

## 13. Common UI Conventions

- **Browse = load.** Selecting a file/folder (or pressing Enter in a path field) loads it immediately; there are no separate “Load” buttons. HDF5 dataset / frame-index changes reload automatically.
  - **No accidental scroll changes.** Spin boxes and drop-downs ignore the mouse wheel — values change only by clicking/typing; the wheel scrolls the panel instead.
  - **Readable right-click menus.** pyqtgraph plot context menus use the dark theme.
  - **Dark theme, orange accent** (Dioplas-inspired); off-white text, light input fields.
  - **Sample-to-detector distance is entered/shown in mm** (Data Viewer, Calibrate seed, Preferences ▶ Geometry). Internally — all calculations — and in written calibration files (paramstest.txt, calibration.json) it is always in **microns**; the mm↔μm conversion happens only at the display boundary. The config key remains `lsd_um` (μm).
  - **Image viewer color scale (Log/cmap/vmin%/vmax%),** shared by the Data Viewer, Calibrate, and Mask Builder image viewers (including Hydra mode): the colorbar’s own zoom defaults to the vmin%/vmax% percentile window rather than the full data range, which a single bad pixel can otherwise stretch into an unreadable sliver. The percentile calculation excludes exact-zero pixels, so a mostly-empty frame (e.g. the Hydra Composite view’s unfilled canvas background) doesn’t skew the window toward looking washed out. Dragging the LUT region or zooming/panning the histogram’s own axis is remembered and reapplied on redraw instead of resetting to the percentile defaults, and toggling Log/Linear converts a manually-set window into the other scale so it keeps pointing at the same data (rather than the same raw numbers) — and always reframes the histogram’s own visible window tightly around the converted levels rather than carrying a manually zoomed/panned window through the same nonlinear conversion, which would otherwise leave the sliders technically in range but squeezed into a barely-visible sliver of the window. Loading new data — a new file, a different frame/dataset, a projection, or a correction/mask change — resets to the percentile defaults; only frame-to-frame updates within an active live stream (Data Viewer’s Live Data card) keep a manual window fixed, matching the live-streaming behavior described in §3.
- 

## 14. Packaging, Deployment & Diagnostics

**Packaging.** `midas-gui` is a MIDAS-style package: `pyproject.toml` (BSD-3-Clause), a `tests/` smoke suite, and `release.sh` for cutting versioned releases (see `RELEASING.md`). Once published it can join the `midas-suite` meta-package as an optional `gui` extra (`pip install midas-suite[gui]`).

**Deployment on another system.** Clone the repo, create the conda environment (`environment.yml`), and run `midas-gui`. The MIDAS analysis backends are installed via the `pip:` section of `environment.yml` (editable from a local MIDAS checkout, from a git subdirectory, or from a private index). `midas_gui/_paths.py` is an inert stub in an installed package (no `sys.path` manipulation).

**Crash diagnostics.** On startup the app installs a logging exception hook and `faulthandler`; any uncaught Python exception or native fault is written to `~/midas_gui_error.log` and shown in a dialog. Installing the hook also prevents PyQt5 from hard-aborting on an exception inside a slot. Each tab is built in isolation — a tab that fails becomes an error placeholder instead of taking the window down. If the app ever “pops up and dies” (typically on Windows), send that log file.

---

## 15. Configuration & Defaults

Every default in the GUI — detector geometry, data/output paths, ring-simulation **materials**, **calibrants**, the **pixel-preset** and **K-edge** menus, the calibration / integration **algorithms**, and **which tabs are visible** — can be set **without editing code**, via a single per-user JSON config that overrides the shipped built-in defaults.

### Where it lives

One per-user file, auto-located per OS: `~/.config/midas_gui/config.json` (Linux), `~/Library/Application Support/midas_gui/config.json` (macOS), `%APPDATA%\midas_gui\config.json` (Windows). If it’s absent, the shipped built-in

defaults are used. Sharing between machines/users is done by exporting/importing a JSON file (below) — copy it wherever you like.

## Profiles — header dropdown + Settings ▶ Preferences... ▶ Profile row

A large **Profile:** [combo ▼] control sits at the top-left of the main window, in the same row as the tab bar (a vertical rule separates it from the tabs) — the fastest way to switch profiles day-to-day. The mirror-image top-right corner of that same row shows the active **Project** name (bold, high-contrast green) whenever File ▶ Project has one open — see §17 “File ▶ Project (FAIR provenance)”. Picking a different profile there switches instantly (no confirmation prompt): live `DEFAULT_*` globals reload, tab visibility re-applies, Hydra mode’s availability updates, and every option-list dropdown/menu backed by that profile (Data Viewer’s Live PV device list, Calibrate’s Calibrant dropdown, and the pixel-size-preset / K-edge-foil popup menus) repopulates from the newly-active profile immediately (see §3 “Mode ribbon”). Profile *management* — creating, duplicating, renaming, deleting — still lives in the Preferences dialog’s own Profile row — `Profile:` [combo ▼] [New...] [Duplicate...] [Rename...] [Delete] — which stays in sync with the header dropdown either way you switch: - Each profile is its own config file under `<config dir>/profiles/<name>.json`; the active one is remembered in `<config dir>/profile_meta.json`. Existing single-config installs are migrated transparently into a profile named **Default** the first time this runs — no data is lost. - Three beamline device presets ship bundled and appear in the combo alongside **Default: 20-ID-D, 20-ID-E, 1-ID-E** — each differs only in its **Devices** list (below), so picking one just swaps the Live Data PV dropdown’s detectors for that beamline’s. They’re seeded once, the first time the app runs on a machine; deleting one doesn’t bring it back. Fresh installs still start on **Default** (same detectors as 20-ID-D) so existing setups are unaffected. - **New...** seeds a blank profile from the shipped built-in defaults. - **Duplicate...** seeds a new profile from whatever is currently shown in the dialog (including unsaved edits), then switches to it. - **Rename...** / **Delete** — Delete refuses to remove the last remaining profile; deleting the active profile switches to another one first. - Switching profiles (via the combo) prompts to confirm if the dialog has unsaved edits, then reloads the dialog’s fields and the live `DEFAULT_*` globals from the new profile. Option lists — the Live Data device dropdown, the Calibrant dropdown, and the pixel-size-preset / K-edge-foil menus — refresh immediately everywhere they’re shown. Seeded default *values* (wavelength, pixel size, Lsd, beam-centre, viewer step sizes, default file paths) only take effect on fields built after the switch — an already-open tab keeps whatever it was showing, since a profile switch should never silently overwrite a value you’re in the middle of editing; a restart applies them everywhere. - The “Local config: ...” label under the row becomes “**Profile ‘<name>’:** `<path>`” so you always know which file you’re editing.

## Editing — Settings ▶ Preferences...

The **Settings** menu opens **Preferences...**, a dialog whose tabs are **pre-filled with the full shipped defaults** so you edit from a complete starting point: - **Geometry** —  $\lambda$ , pixel size, Lsd, beam centre. - **Data Viewer** — the up/down-arrow step size for each field in the Data Viewer’s Ring simulation card ( $\lambda$ , max 2 $\theta$ , Lsd, pixel size, BC<sub>y</sub>/BC<sub>z</sub>, ty/tz). Shipped defaults: 0.01 Å, 1°, 1 mm, 0.1  $\mu$ m, 1 px, 0.1° respectively. - **Paths** — default data / calibration / output files & folders. - **Materials / Calibrants** — add / remove / modify (name + lattice + SG). - **Devices** — the detector devices offered in the Data Viewer’s **Live Data** PV dropdown (name, prefix, PVA suffix). The live PV is built as `prefix + PVA suffix`. All PVA suffixes are `Pva1:Image`, plus a built-in **Sim Detector** entry (`midasSim:` prefix) for hardware-free testing — see the Live Data card section above; add / remove / edit rows for your own beamline’s devices, or switch to one of the bundled beamline **Profiles** above instead of hand-editing: - **20-ID-D** — `20iddNF (20idOR1:), s20idPil (20idPil:), pg4 (1idPG4:), 20iddTomo (20idGH1s:), 20iddFF (20IDFF:)`. - **20-ID-E** — `pimega (PITEC:D:RAD1_5Mh:), spl1 (20idsp1:), s20varex2 (20idVarex2:), pg6 (20idPG6s:), gh2 (20idGH2S:)`. - **1-ID-E** — `ge1-ge5 (GE1:-GE5:), pixirad (s1_pixirad2:), gh1 (1idGH1:), pg1 (1idPG1:), pg5 (1idSP5:), s1varex1 (1idVarex1:)`. Names match each detector’s variable name in B-PILOT (`mpe_bluesky/instrument/devices/`), the beamline’s Bluesky/ophyd device definitions, so entries are traceable back to source. - **Menus** — the pixel-size presets and K-edge foils. - **Algorithms** — default calibration pipeline, integration kernel, output format, error model, colormap/theme. - **Tabs** — which tabs are visible. Data Viewer, Mask Builder, Calibrate and Batch Integrate are always shown (their boxes are locked on); tick/untick the rest. **Tab visibility applies immediately** on save (no restart), unlike the other settings. - **Display** — the **interface scale**, a whole-application zoom (layout *and* fonts) for HiDPI / 4K monitors where the default looks too small. Pick a value or a preset (100 %  $\approx$  1080p, 150 %  $\approx$  1440p, 200 %  $\approx$  4K); it applies after a restart (you’re offered one on save). Also reachable from **Settings ▶ Interface scaling...** It is implemented with Qt’s `QT_SCALE_FACTOR`, so the whole layout scales uniformly.

Buttons: **Save as my defaults** (write your config), **Save current GUI state** (capture the Data Viewer’s live  $\lambda$ /pixel/Lsd/BC into the fields), **Save config to JSON... / Load config (JSON)...** (export/import a config to share), and **Reset to shipped defaults** (delete your config). Also on the menu: **Open config folder** and **Reload config**. Saving writes your per-user file; **changes take effect on the next launch**.

## File format

```
{
  "geometry": { "wavelength_A": 0.39, "pixel_um": 75.0, "lsd_um": 121000.0,
```



```

        "bc_y": 10.0, "bc_z": 10.0,
        "pixel_presets": [{"Eiger", 75.0}, {"Pilatus", 172.0}],
        "k_edge_foils": [{"Au", 80.725}, {"Pb", 88.005}],
    "viewer_steps": { "wavelength": 0.01, "two_theta": 1.0, "lsd_mm": 1.0,
        "pixel": 0.1, "bc": 1.0, "tilt": 0.1 },
    "materials": { "Ni (FCC)": {"a":3.5238,"b":3.5238,"c":3.5238,"alpha":90,"beta":90,"gamma":90,"sg":225} },
    "calibrants": { "CeO2": {"a":5.4116,"b":5.4116,"c":5.4116,"alpha":90,"beta":90,"gamma":90,"sg":225} },
    "devices": [ {"name": "s20varex1", "prefix": "20IDFF:", "pva_suffix": "Pva1:Image"} ],
    "paths": { "nickel_h5": "/data/mygroup/sample.h5", "calib_file": "/data/mygroup/calibration.json" },
    "ui": { "calibration_pipeline": "one_shot", "integration_kernel": "subpixel2",
        "output_format": "csv", "azimuthal_method": "poisson", "plot_theme": "hot",
        "visible_tabs": ["Calib. Refinement", "Pump Probe"],
        "ui_scale": 1.0 }
}

```

- `ui.visible_tabs` lists the **optional** tabs to show (the four always-on tabs are implicit). The shipped default is `["Calib. Refinement", "Pump Probe"]` — Corrections, PDF Analysis, Texture and Results & Export are hidden until you add them. Edit it from **Preferences ▶ Tabs**.
- `ui.ui_scale` is the whole-interface zoom (0.5–4.0) applied at startup via `QT_SCALE_FACTOR`; ~1.5 for 1440p, ~2.0 for 4K. Edit it from **Preferences ▶ Display or Settings ▶ Interface scaling...** (takes effect on restart).
- When a **list section is present it fully replaces** that built-in list — so `materials`, `calibrants`, `devices`, `pixel_presets` and `k_edge_foils` in the file are your complete lists. (The Preferences dialog pre-fills them with the shipped entries, so you always start from the full set; use **Reset to shipped defaults** to get them back.) `sg` is the space-group number.
- Geometry scalars, `paths` and `ui` are simple overrides; any section or key may be omitted. A malformed config is ignored (built-ins are used) rather than blocking startup.
- A minimal template lives at `documentation/config.example.json`; the step-by-step guide is `documentation/config_gui.md`.

## 16. File ▶ Save/Load GUI State

Beyond a saved **profile** of defaults (§15), the **File** menu can capture the **live, in-progress state of every tab** — every field you’ve typed or picked, across all 10 tabs — to one JSON file, so a session can be closed and resumed later exactly where it left off.

- **File ▶ Save GUI State...** (`Ctrl+S`) — the first time in a session, prompts for a destination (default `midas_session.json`); every subsequent `Ctrl+S` **silently overwrites that same file** (no dialog) as long as the session hasn’t loaded/saved a different file since. **File ▶ Save GUI State As...** (`Ctrl+Shift+S`) always prompts for a destination, and makes *that* file the target for future plain `Ctrl+S`. Either way, a completion dialog lists which tabs (if any) failed to save.
- **File ▶ Load GUI State...** (`Ctrl+O`) — asks for confirmation (loading overwrites every tab’s current values), then pick a previously saved file. A completion dialog reports any tabs present in the file that no longer exist, or that failed to restore. The tab that was active at save time is re-selected.

### What is restored automatically

Every field (text boxes, spin boxes, checkboxes, combo/dropdown selections) in every tab is restored as typed. In addition, **path-backed data is reloaded from disk**, the same way it loads when you type/browse to a path by hand — images, masks-by-path, dark/bright/background frames, and HDF5 datasets all re-read their file automatically after a state load (guarded so a moved/deleted file is skipped quietly rather than popping a warning).

### What is *not* re-run

Loading a state **does not re-run any long-running pipeline** — Calibrate’s Fit, Batch Integrate, the PDF transform, and Calibration Refinement all keep their inputs restored but require **one manual click of the tab’s own Run/Fit button** to reproduce their result. This keeps a load fast and avoids silently kicking off a multi-minute job in the background.

### Sidecar files for in-progress derived data

Two tabs can hold computed data that hasn’t been exported to a file of its own yet — a drawn/computed **mask** (Mask Builder) and a just-fit **calibration result** (Calibrate). Saving a GUI state writes small sidecar files next to it so this in-progress work isn’t silently lost: - `<state file stem>_mask.tif` — the current in-memory mask, if any; reloaded automatically on the next Load. - `<state file stem>_calibration.json` — the last fit result’s flattened fields, kept for the record and to reseed the manual/seed

geometry fields. This does **not** reconstruct the in-memory fitted result object (it isn't a plain, re-loadable structure) — re-run **Fit** after loading to reproduce it.

---

## 17. File ▶ Project (FAIR provenance)

Separate from GUI State (§16, which snapshots widget *configuration* so a session can be resumed), a **Project** is a long-lived record of what actually *happened*: every time a Calibrate or Batch Integrate run finishes, a self-contained record is appended to it — enough to reproduce that exact run later (inputs, parameters, results, and software versions). It's designed to accumulate across many separate launches of the GUI over the course of an experiment, not just one session.

- **File ▶ New Project...** — pick a destination `.h5` file; it's created empty and becomes the active project immediately. Refuses to overwrite an existing file (use **Open Project...** to continue one).
- **File ▶ Open Project...** — pick a previously-created project file to make it active. If it has any recorded attempts, a **Populate from project** dialog follows immediately — see below.
- **File ▶ Close Project** — stops logging; no data already written is touched.
- The active project's filename is always shown in the status bar (bottom-right, "Project: ..." / "Project: none") **and**, in a bold, high-contrast green, at the far right of the tab-bar header row (empty when no project is open) — the header indicator is the one meant to catch your eye; the status-bar one is the quiet, always-present record.

### What gets logged, and when

While a project is open, **every completed Calibrate run** (single-detector Tab 2, or each of the 4 panels in Hydra mode) and **every completed Batch Integrate run** (single-detector Tab 4, or each Hydra panel) appends one record — automatically, with no extra action needed. Nothing is logged when no project is open (the default), so this is entirely opt-in.

Each record is self-contained and includes: the exact parameters used (pipeline/refine choices for a calibration, kernel/binning/corrections for an integration); the fitted calibration result or the integrated profile(s); the mask and dark/bright/background frames actually applied (embedded, since a mask drawn directly in Mask Builder has no file of its own to point back to); file paths and checksums for the raw calibrant image / raw scan data referenced (not duplicated — a scan can be many thousands of frames); and the midas-gui / MIDAS package versions active at the time. A Batch Integrate record additionally embeds the exact calibration values it used and links back to the specific Calibrate run that produced them, when that run was itself logged to the same project.

Records are never overwritten — recalibrating or re-integrating adds a new, separately numbered record rather than replacing the previous one, so the full history of what was tried stays in the file. A failed provenance write (e.g. disk full) is logged to the tab's own Log panel and never blocks or alters the run's on-screen result.

The file is plain HDF5 ( `h5py` / `h5dump` /HDFView can browse it directly) — no separate tool is required to inspect what a project contains.

### Opening a project can populate the GUI

**File ▶ Open Project...** doesn't just make a project active for future logging — if it contains any recorded attempts, a **Populate from project** dialog appears with one row per panel found in the file (**Single detector**, or each of **ge1–ge4** present in Hydra mode). Each row has two independent checkboxes, **Calibrate** and **Batch Integrate**, each paired with a picker defaulting to that panel's latest attempt (older attempts stay selectable if the project has more than one). A checkbox is disabled if that panel has no attempt of that kind. Click **Populate** to apply the checked rows, or **Skip** to leave every tab exactly as it was (the project still becomes active for future logging either way).

Populating switches the Calibrate/Batch Integrate mode ribbon (Single-detector vs Hydra) to match what was selected, and restores:

- **Calibrate**: the data path/dataset (and, single-detector only, the frame index), wavelength, pixel size, calibrant, refine-parameter checkboxes, iteration counts, device, and the fitted BC/Lsd/tilts as a manual seed (**Use manual seed** is turned on) — a single click of **Fit** reproduces the recorded result.
- **Batch Integrate**: the data path/dataset and frame range, kernel, output format, and monitor file — plus, unlike a plain GUI-state load, a live, ready-to-use **calibration** built from the attempt's own recorded geometry, so **Run** works immediately without first re-running Tab 2.

Populating also redraws the recorded run's **results**, immediately, without needing to press Fit or Run again:

- **Calibrate:** the predicted-ring overlay reappears on the image at once (pure geometry — no image needed). If the attempt's data file can still be loaded, the **Radial Profile** and **Eta vs R Cake** tabs are also recomputed and shown right away, exactly as if Fit had just finished. In Hydra mode this happens per panel — switch panels (or the panel is the toolbar's current one already) to see each one's rings.
- **Batch Integrate:** the **Waterfall** and **Stacked profiles** views are replayed from the attempt's own saved per-frame profiles — no re-run needed. In Hydra mode, each GE panel keeps its own independent pair of views, so switching the **GE1/GE2/GE3/GE4** toolbar shows that panel's own recorded run, not a shared plot.

Only fields the codebase can already restore from a file path are populated this way — dark/bright/background/mask sources that were embedded directly in the project (no file of their own, e.g. a mask hand-drawn in Mask Builder) are not re-created automatically and must be re-added from Mask Builder / the Data card if needed.

---

*For bugs or questions, see the MIDAS GUI repository.*